

Jordan 5G SA – Master Deployment Plan

10 rollouts · 36 months · 6,270 macro sites · 18,675 sectors · ≈ 8,891 km fiber · 51 indoor venues

Programme month 1 (M1) is assumed to be **January 2027**; all dates shift with the actual contract-effective date (START in `tools/deployment_plan.py`). This plan implements the nominal design in `5G_Jordan_Design_Report.md`; all quantities are generated from it and regenerate with `python tools/deployment_plan.py && python tools/deployment_docs.py`.

Document set

#	Document	Purpose
00	Master Deployment Plan (this file)	Strategy, sequencing, schedule, governance
01	<code>01_Programme_Schedule.md</code>	Month-by-month schedule, milestones, stage gates
02	<code>02_Transport_and_Core_Deployment.md</code>	Core, backbone, hub and ring build sequence
03	<code>03_Site_Implementation_Standard.md</code>	Site process from survey to acceptance, checklists, 1 Gbps acceptance test
04	<code>04_Indoor_Deployment_Plan.md</code>	Venue programme by rollout
05	<code>05_Resource_and_Procurement_Plan.md</code>	Crews, lead times, logistics, call-off calendar
06	<code>06_Risk_Register.md</code>	Risks, owners, mitigations
LLD	<code>../lld/LLD-01...03</code>	Equipment-level design: RAN, core, transport (ports, DWDM, microwave)
RO	<code>RO-01 ... RO-10_Work_Package.md</code>	One work package per rollout: scope, BoQ, prerequisites, dates, exit criteria
Data	<code>Jordan_5G_Deployment_Plan.xlsx</code> , <code>rollout_sites.csv</code> , <code>rollout_boq.csv</code> , <code>rollout_schedule.csv</code> , <code>rollout_venues.csv</code>	Site-level allocation, bill of quantities, Gantt
Map	<code>index.html</code> → layer “Rollout phase” + rollout slider	Cumulative build on the satellite map

1. Deployment strategy

- Build unit = one hub cluster** (DU hotel / pre-aggregation hub with all its access rings, ≈ 40 sites). Fiber, hub and radio sites of a unit are always in the same rollout, so no site waits for transport and no ring is left open.
- Value first, inside-out.** Units are ranked by service class (dense urban > urban > suburban), strategic weight (Amman–Zarqa metro, Irbid, Aqaba gateway, Dead Sea and Petra tourism) and distance from the city centre. Each city therefore grows as a contiguous 1 Gbps island – no isolated sites with a poor-SINR edge.
- Transport leads radio.** A unit cannot start before the backbone route feeding its city is lit (see 02). Highway sites go on air one rollout after their route, using the backbone add/drop points.
- Ramp, plateau, taper.** 306 → 465 → 617 sites in the first three rollouts while teams, logistics and processes mature, a plateau of ≈ 720–770 sites per rollout (≈ 250 per month) through RO-04...RO-08, taper in RO-09/10 when work moves to distant, low-density areas.
- Rural and highway coverage is not left to the end.** National highway corridors go on air one rollout after their backbone route (from RO-04); rural packages start in RO-07 so coverage obligations are met before

programme close.

6. **Indoor follows macro by one rollout** so the donor/neighbour layer exists when a venue is integrated; landmark venues (QAIA, Abdali, major malls and 5-star hotels) start in RO-02.
7. **Every rollout ends with a formal cluster acceptance** against the 1 Gbps KPI (03 §6) before commercial launch in that area.

2. The ten rollouts

Rollout	On-air window	Sites	DU	U	SU	Rural	Hwy	Hubs	Access + agg fiber km	Backbone km	Indoor venues	Cum. sites	Cum. pop. covered	Cum. 1 Gbps km ²	Main areas
RO-01	Apr 2027 – Jun 2027	306	293	13	0	0	0	6	260	87	0	306	8.6 %	40	Amman-Zarqa Metro
RO-02	Jul 2027 – Sep 2027	465	176	289	0	0	0	10	445	194	18	771	18.4 %	117	Amman-Zarqa Metro, Irbid
RO-03	Oct 2027 – Dec 2027	617	8	589	20	0	0	15	645	340	5	1,388	28.5 %	236	Amman-Zarqa Metro, Aqaba, Irbid
RO-04	Jan 2028 – Mar 2028	719	0	625	61	0	33	16	719	337	14	2,107	39.3 %	376	Amman-Zarqa Metro, Irbid, Aqaba, Desert Highway (R15) Amman-Aqaba
RO-05	Apr 2028 – Jun 2028	769	0	561	176	0	32	19	806	278	1	2,876	50.1 %	550	Amman-Zarqa Metro, Wadi Musa / Petra, Irbid, Dead Sea - Wadi Araba (R65) Amman-Aqaba
RO-06	Jul 2028 – Sep 2028	770	0	455	308	0	7	19	916	174	9	3,646	60.3 %	755	Irbid, Amman-Zarqa Metro, Madaba, Mafrq
RO-07	Oct 2028 – Dec 2028	766	0	264	191	311	0	13	516	185	0	4,412	73.9 %	878	Jerash, Ma'an, Salt, Mafrq

Rollout	On-air window	Sites	DU	U	SU	Rural	Hwy	Hubs	Access + agg fiber km	Backbone km	Indoor venues	Cum. sites	Cum. pop. covered	Cum. 1 Gbps km ²	Main areas
RO-08	Jan 2029 – Mar 2029	718	0	166	415	126	11	18	774	471	2	5,130	83.4 %	1,071	Amman-Zarqa Metro, Ajloun, Amman governorate (rural), Salt
RO-09	Apr 2029 – Jun 2029	669	0	2	609	7	51	17	885	0	0	5,799	88.9 %	1,309	Amman-Zarqa Metro, Irbid, Mafraq-Safawi-Ruwaished (R10), Zarqa-Azraq-Safawi
RO-10	Jul 2029 – Sep 2029	471	0	6	465	0	0	17	659	0	2	6,270	93.0 %	1,491	Amman-Zarqa Metro, Irbid, Mutah-Mazar, Aqaba

Cumulative population = share of Jordan's population living inside the footprint built so far (design-model estimate).

Year 1 (2027, RO-01...RO-03): 1,388 sites. Core live in M3–M4; all of dense-urban Amman and Irbid core, inner urban Amman, Aqaba city; northern backbone ring and Desert Highway lit. ≈ 28.5 % of population covered. **Year 2 (2028, RO-04...RO-07): 3,024 sites.** Remaining urban Amman, Zarqa, Russeifa, Irbid; most governorate capitals (Ajloun and the last Salt/Madaba clusters follow in RO-08); Dead Sea and Petra; southern backbone ring closed; national highways; first rural packages. ≈ 73.9 %. **Year 3 (2029, RO-08...RO-10): 1,858 sites.** Suburban rings, remaining rural, eastern desert corridors, n258 hotspot layer in dense urban, indoor phase completion, network-wide optimisation and hand-over. 93.0 %.

3. Timeline

(Gantt chart: see workbook [Jordan_5G_Deployment_Plan.xlsx](#), sheet "Schedule (Gantt)", and the activity table in document 01.)

Each rollout runs the same 10-month pipeline, overlapping with its neighbours (three to four rollouts are always in flight):

Offset from on-air start (T)	Activity
T-6 ... T-3	Nominal release, technical site surveys (TSSR), acquisition, permits
T-5 ... T-3	Detailed RF and transport design freeze, material call-off
T-4 ... T-1	Civil works, power, fiber to hubs and rings
T-2 ... T	Hub / DU-hotel and router commissioning
T-1 ... T+2	Radio installation, commissioning, integration to 5GC

Offset from on-air start (T)	Activity
T ... T+2	Single-site verification
T+1 ... T+4	Cluster optimisation and acceptance
T+3 ... T+4	Commercial launch of the area, hand-over to operations

RO-01 is compressed (acquisition starts in M1): it must use **existing rooftops and towers under sharing agreements** – this is the first critical-path item.

4. Critical path and prerequisites

#	Prerequisite	Needed by	Owner
1	Spectrum licence: n78 3×100 MHz, n28, n1/n3, n258; type approval	M1 (RO-01 radios ordered against exact carrier plan)	Regulatory
2	RAN / core / transport vendor contracts, frame pricing, delivery SLAs	M1	Procurement
3	Tower-company / operator site-sharing and dark-fiber framework agreements	M1–M2	Commercial
4	DC1 and DC2 ready (power, cooling, cloud platform), 5GC + IMS integrated, first call	M3–M4	Core
5	Amman metro core ring and first 6 hubs lit	M3	Transport
6	Municipality (GAM, ASEZA, municipalities) master permit / fast-track process, EMF compliance procedure	M2	Site acquisition
7	Power utility (JEPCO / IDECO / EDCO) connection framework	M2	Civil
8	Northern backbone ring (RO-02), Desert Highway (RO-03), Wadi Araba – Dead Sea (RO-04)	start of the named rollout	Transport
9	Cross-border coordination for 109 flagged sectors	before the affected rollout goes on air	Regulatory
10	3-CC n78 capable devices / CPE in the market	commercial launch RO-01 (M7)	Marketing

5. Governance

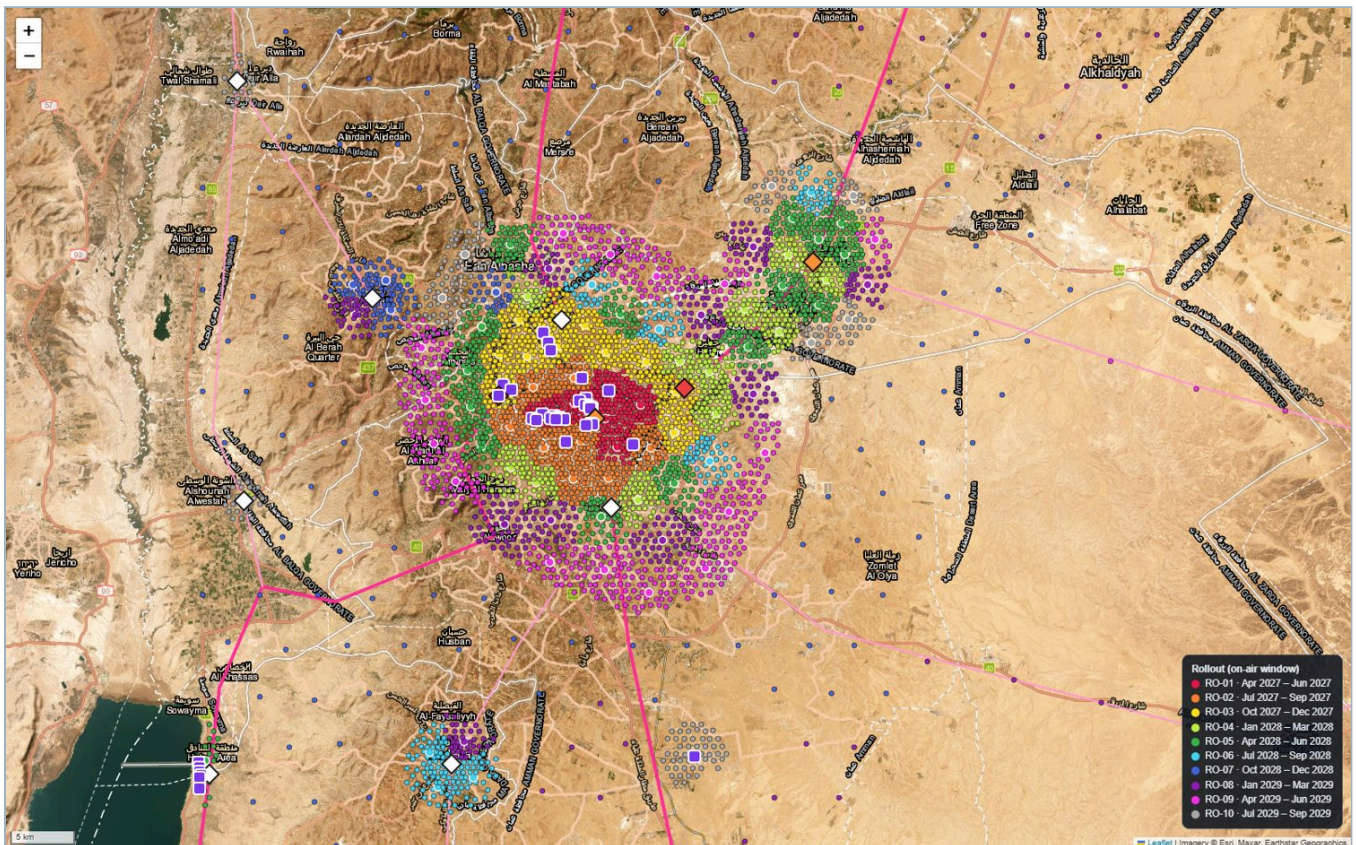
- **Programme board** (monthly): sponsor, CTO, CFO, vendor executives – budget, stage gates, scope changes.
- **PMO** (weekly): rollout managers (one per active rollout), streams for Acquisition, Civil, Transport, RAN, Core, Indoor, Optimisation, HSE/Quality, Logistics.
- **Stage gates per rollout**: G1 design freeze (T–3) · G2 ready-for-installation ≥ 80 % sites (T–1) · G3 on-air ≥ 95 % (T+3) · G4 cluster acceptance (T+4) · G5 hand-over to operations.
- **RACI (summary)**: operator – accountable for spectrum, acquisition approvals, acceptance; RAN vendor – responsible for install, commissioning, SSV, optimisation; transport contractor – fiber and routers; tower-co – passive infrastructure; PMO – integrated schedule and reporting.
- **Reporting KPIs**: sites acquired / RFI / installed / on-air / accepted vs plan (weekly S-curve per rollout), fiber km built, first-time-right install rate (target ≥ 92 %), SSV pass rate, % drive-test bins ≥ 1 Gbps, LTIFR, material availability ≥ 98 % at call-off.

6. Peak resources (see 05)

Peak in RO-05...RO-07: ≈ 37 radio install crews, 278 civil crews, 78 urban fiber crews, 22 acquisition agents, 5 SSV teams and 6 optimisation teams; peak build rate ≈ 257 sites/month and ≈ 310 km of access fiber per month.

7. Planning assumptions

- $\geq 50\%$ of urban sites reuse existing rooftops/towers (sharing) – without this the acquisition lead time moves from 3 to 6–9 months and RO-01...RO-03 slip by one quarter.
- Backbone: leased / swapped dark fiber (existing national OPGW and operator routes) is acceptable for first light; own-build follows within the same rollout year.
- Working calendar 22 days/month; Ramadan and summer-heat productivity factor 0.85 applied to Q2 rollouts in the resource plan buffer.
- Quantities come from a nominal design: expect $\pm 10\%$ on site count after surveys; each work package is re-baselined at gate G1.



Rollout sequence, Amman-Zarqa metro: colour = rollout in which the hub cluster goes on air.

01 – Programme Schedule (36 months, 10 rollouts)

(Gantt chart: see workbook [Jordan_5G_Deployment_Plan.xlsx](#), sheet "Schedule (Gantt)", and the activity table in document 01.)

Month-by-month milestones

Month	Date	Milestones
M1	Jan 2027	Contract effective; PMO mobilised; RO-01 acquisition starts; long-lead orders placed; RO-01 gate G1 design freeze, material call-off
M2	Feb 2027	Sharing and dark-fiber frameworks signed; warehouse operational (Amman)
M3	Mar 2027	DC1/DC2 ready; metro core ring lit; first hubs commissioned
M4	Apr 2027	5GC + IMS first call; first site on air ; RO-01 on-air window opens (306 sites); RO-02 gate G1 design freeze, material call-off; RO-03 nominal release, surveys and acquisition start
M5	May 2027	–
M6	Jun 2027	–
M7	Jul 2027	Commercial launch – Amman core (RO-01 accepted); RO-01 gate G3 (≥ 95 % on air); RO-02 on-air window opens (465 sites); RO-03 gate G1 design freeze, material call-off; RO-04 nominal release, surveys and acquisition start
M8	Aug 2027	RO-01 cluster acceptance G4 / area launch
M9	Sep 2027	–
M10	Oct 2027	RO-02 gate G3 (≥ 95 % on air); RO-03 on-air window opens (617 sites); RO-04 gate G1 design freeze, material call-off; RO-05 nominal release, surveys and acquisition start
M11	Nov 2027	RO-02 cluster acceptance G4 / area launch
M12	Dec 2027	Year-1 close: Amman/Irbid dense urban + Aqaba on air; Desert Highway backbone lit
M13	Jan 2028	RO-03 gate G3 (≥ 95 % on air); RO-04 on-air window opens (719 sites); RO-05 gate G1 design freeze, material call-off; RO-06 nominal release, surveys and acquisition start
M14	Feb 2028	RO-03 cluster acceptance G4 / area launch
M15	Mar 2028	Southern backbone ring closed (Wadi Araba – Dead Sea)
M16	Apr 2028	RO-04 gate G3 (≥ 95 % on air); RO-05 on-air window opens (769 sites); RO-06 gate G1 design freeze, material call-off; RO-07 nominal release, surveys and acquisition start
M17	May 2028	RO-04 cluster acceptance G4 / area launch
M18	Jun 2028	Half-way: ≥ 50 % population covered
M19	Jul 2028	RO-05 gate G3 (≥ 95 % on air); RO-06 on-air window opens (770 sites); RO-07 gate G1 design freeze, material call-off; RO-08 nominal release, surveys and acquisition start
M20	Aug 2028	RO-05 cluster acceptance G4 / area launch

Month	Date	Milestones
M21	Sep 2028	–
M22	Oct 2028	RO-06 gate G3 ($\geq 95\%$ on air); RO-07 on-air window opens (766 sites); RO-08 gate G1 design freeze, material call-off; RO-09 nominal release, surveys and acquisition start
M23	Nov 2028	RO-06 cluster acceptance G4 / area launch
M24	Dec 2028	Year-2 close: urban layer of all major cities on air, rural programme started
M25	Jan 2029	RO-07 gate G3 ($\geq 95\%$ on air); RO-08 on-air window opens (718 sites); RO-09 gate G1 design freeze, material call-off; RO-10 nominal release, surveys and acquisition start
M26	Feb 2029	RO-07 cluster acceptance G4 / area launch
M27	Mar 2029	–
M28	Apr 2029	n258 hotspot layer starts (dense urban); RO-08 gate G3 ($\geq 95\%$ on air); RO-09 on-air window opens (669 sites); RO-10 gate G1 design freeze, material call-off
M29	May 2029	RO-08 cluster acceptance G4 / area launch
M30	Jun 2029	–
M31	Jul 2029	RO-09 gate G3 ($\geq 95\%$ on air); RO-10 on-air window opens (471 sites)
M32	Aug 2029	RO-09 cluster acceptance G4 / area launch
M33	Sep 2029	Last site on air
M34	Oct 2029	RO-10 gate G3 ($\geq 95\%$ on air)
M35	Nov 2029	RO-10 cluster acceptance G4 / area launch
M36	Dec 2029	Final acceptance, as-built documentation, hand-over to operations; programme close

S-curve (cumulative sites on air at end of each window)

Rollout	End of window	Cumulative sites	% of programme	Cum. population covered
RO-01	Jun 2027	306	5 %	8.6 %
RO-02	Sep 2027	771	12 %	18.4 %
RO-03	Dec 2027	1,388	22 %	28.5 %
RO-04	Mar 2028	2,107	34 %	39.3 %
RO-05	Jun 2028	2,876	46 %	50.1 %
RO-06	Sep 2028	3,646	58 %	60.3 %
RO-07	Dec 2028	4,412	70 %	73.9 %
RO-08	Mar 2029	5,130	82 %	83.4 %
RO-09	Jun 2029	5,799	92 %	88.9 %
RO-10	Sep 2029	6,270	100 %	93.0 %

Activity schedule per rollout

Rollout	Activity	Start	End
RO-01	Nominal release, TSSR surveys, site acquisition & permits	M1 Jan 2027	M1 Jan 2027
RO-01	Detailed RF / transport design freeze, material call-off	M1 Jan 2027	M1 Jan 2027
RO-01	Civil works, power, fiber build to hubs and rings	M1 Jan 2027	M3 Mar 2027
RO-01	Hub / DU-hotel and router commissioning	M2 Feb 2027	M4 Apr 2027
RO-01	Radio install, commissioning, integration to 5GC	M3 Mar 2027	M6 Jun 2027
RO-01	Single-site verification (SSV)	M4 Apr 2027	M6 Jun 2027
RO-01	Cluster optimisation & acceptance (1 Gbps drive test)	M5 May 2027	M8 Aug 2027
RO-01	Commercial launch of rollout area / hand-over to operations	M7 Jul 2027	M8 Aug 2027
RO-02	Nominal release, TSSR surveys, site acquisition & permits	M1 Jan 2027	M4 Apr 2027
RO-02	Detailed RF / transport design freeze, material call-off	M2 Feb 2027	M4 Apr 2027
RO-02	Civil works, power, fiber build to hubs and rings	M3 Mar 2027	M6 Jun 2027
RO-02	Hub / DU-hotel and router commissioning	M5 May 2027	M7 Jul 2027
RO-02	Radio install, commissioning, integration to 5GC	M6 Jun 2027	M9 Sep 2027
RO-02	Single-site verification (SSV)	M7 Jul 2027	M9 Sep 2027
RO-02	Cluster optimisation & acceptance (1 Gbps drive test)	M8 Aug 2027	M11 Nov 2027
RO-02	Commercial launch of rollout area / hand-over to operations	M10 Oct 2027	M11 Nov 2027
RO-03	Nominal release, TSSR surveys, site acquisition & permits	M4 Apr 2027	M7 Jul 2027
RO-03	Detailed RF / transport design freeze, material call-off	M5 May 2027	M7 Jul 2027
RO-03	Civil works, power, fiber build to hubs and rings	M6 Jun 2027	M9 Sep 2027
RO-03	Hub / DU-hotel and router commissioning	M8 Aug 2027	M10 Oct 2027
RO-03	Radio install, commissioning, integration to 5GC	M9 Sep 2027	M12 Dec 2027
RO-03	Single-site verification (SSV)	M10 Oct 2027	M12 Dec 2027
RO-03	Cluster optimisation & acceptance (1 Gbps drive test)	M11 Nov 2027	M14 Feb 2028
RO-03	Commercial launch of rollout area / hand-over to operations	M13 Jan 2028	M14 Feb 2028
RO-04	Nominal release, TSSR surveys, site acquisition & permits	M7 Jul 2027	M10 Oct 2027
RO-04	Detailed RF / transport design freeze, material call-off	M8 Aug 2027	M10 Oct 2027
RO-04	Civil works, power, fiber build to hubs and rings	M9 Sep 2027	M12 Dec 2027
RO-04	Hub / DU-hotel and router commissioning	M11 Nov 2027	M13 Jan 2028
RO-04	Radio install, commissioning, integration to 5GC	M12 Dec 2027	M15 Mar 2028
RO-04	Single-site verification (SSV)	M13 Jan 2028	M15 Mar 2028
RO-04	Cluster optimisation & acceptance (1 Gbps drive test)	M14 Feb 2028	M17 May 2028
RO-04	Commercial launch of rollout area / hand-over to operations	M16 Apr 2028	M17 May 2028
RO-05	Nominal release, TSSR surveys, site acquisition & permits	M10 Oct 2027	M13 Jan 2028
RO-05	Detailed RF / transport design freeze, material call-off	M11 Nov 2027	M13 Jan 2028
RO-05	Civil works, power, fiber build to hubs and rings	M12 Dec 2027	M15 Mar 2028
RO-05	Hub / DU-hotel and router commissioning	M14 Feb 2028	M16 Apr 2028
RO-05	Radio install, commissioning, integration to 5GC	M15 Mar 2028	M18 Jun 2028
RO-05	Single-site verification (SSV)	M16 Apr 2028	M18 Jun 2028
RO-05	Cluster optimisation & acceptance (1 Gbps drive test)	M17 May 2028	M20 Aug 2028
RO-05	Commercial launch of rollout area / hand-over to operations	M19 Jul 2028	M20 Aug 2028

Rollout	Activity	Start	End
RO-06	Nominal release, TSSR surveys, site acquisition & permits	M13 Jan 2028	M16 Apr 2028
RO-06	Detailed RF / transport design freeze, material call-off	M14 Feb 2028	M16 Apr 2028
RO-06	Civil works, power, fiber build to hubs and rings	M15 Mar 2028	M18 Jun 2028
RO-06	Hub / DU-hotel and router commissioning	M17 May 2028	M19 Jul 2028
RO-06	Radio install, commissioning, integration to 5GC	M18 Jun 2028	M21 Sep 2028
RO-06	Single-site verification (SSV)	M19 Jul 2028	M21 Sep 2028
RO-06	Cluster optimisation & acceptance (1 Gbps drive test)	M20 Aug 2028	M23 Nov 2028
RO-06	Commercial launch of rollout area / hand-over to operations	M22 Oct 2028	M23 Nov 2028
RO-07	Nominal release, TSSR surveys, site acquisition & permits	M16 Apr 2028	M19 Jul 2028
RO-07	Detailed RF / transport design freeze, material call-off	M17 May 2028	M19 Jul 2028
RO-07	Civil works, power, fiber build to hubs and rings	M18 Jun 2028	M21 Sep 2028
RO-07	Hub / DU-hotel and router commissioning	M20 Aug 2028	M22 Oct 2028
RO-07	Radio install, commissioning, integration to 5GC	M21 Sep 2028	M24 Dec 2028
RO-07	Single-site verification (SSV)	M22 Oct 2028	M24 Dec 2028
RO-07	Cluster optimisation & acceptance (1 Gbps drive test)	M23 Nov 2028	M26 Feb 2029
RO-07	Commercial launch of rollout area / hand-over to operations	M25 Jan 2029	M26 Feb 2029
RO-08	Nominal release, TSSR surveys, site acquisition & permits	M19 Jul 2028	M22 Oct 2028
RO-08	Detailed RF / transport design freeze, material call-off	M20 Aug 2028	M22 Oct 2028
RO-08	Civil works, power, fiber build to hubs and rings	M21 Sep 2028	M24 Dec 2028
RO-08	Hub / DU-hotel and router commissioning	M23 Nov 2028	M25 Jan 2029
RO-08	Radio install, commissioning, integration to 5GC	M24 Dec 2028	M27 Mar 2029
RO-08	Single-site verification (SSV)	M25 Jan 2029	M27 Mar 2029
RO-08	Cluster optimisation & acceptance (1 Gbps drive test)	M26 Feb 2029	M29 May 2029
RO-08	Commercial launch of rollout area / hand-over to operations	M28 Apr 2029	M29 May 2029
RO-09	Nominal release, TSSR surveys, site acquisition & permits	M22 Oct 2028	M25 Jan 2029
RO-09	Detailed RF / transport design freeze, material call-off	M23 Nov 2028	M25 Jan 2029
RO-09	Civil works, power, fiber build to hubs and rings	M24 Dec 2028	M27 Mar 2029
RO-09	Hub / DU-hotel and router commissioning	M26 Feb 2029	M28 Apr 2029
RO-09	Radio install, commissioning, integration to 5GC	M27 Mar 2029	M30 Jun 2029
RO-09	Single-site verification (SSV)	M28 Apr 2029	M30 Jun 2029
RO-09	Cluster optimisation & acceptance (1 Gbps drive test)	M29 May 2029	M32 Aug 2029
RO-09	Commercial launch of rollout area / hand-over to operations	M31 Jul 2029	M32 Aug 2029
RO-10	Nominal release, TSSR surveys, site acquisition & permits	M25 Jan 2029	M28 Apr 2029
RO-10	Detailed RF / transport design freeze, material call-off	M26 Feb 2029	M28 Apr 2029
RO-10	Civil works, power, fiber build to hubs and rings	M27 Mar 2029	M30 Jun 2029
RO-10	Hub / DU-hotel and router commissioning	M29 May 2029	M31 Jul 2029
RO-10	Radio install, commissioning, integration to 5GC	M30 Jun 2029	M33 Sep 2029
RO-10	Single-site verification (SSV)	M31 Jul 2029	M33 Sep 2029
RO-10	Cluster optimisation & acceptance (1 Gbps drive test)	M32 Aug 2029	M35 Nov 2029
RO-10	Commercial launch of rollout area / hand-over to operations	M34 Oct 2029	M35 Nov 2029

The colour Gantt is in [Jordan_5G_Deployment_Plan.xlsx](#) (sheet "Schedule (Gantt)").

02 – Transport and Core Deployment Plan

1. 5G core

Phase	Months	Scope
C0	M1–M2	DC1 (Amman West) and DC2 (Amman East) facility readiness, cloud platform, IP fabric, security zones
C1	M2–M4	5GC control plane (AMF, SMF, AUSF, UDM/UDR, PCF, NRF, NSSF, SCP, CHF), UPF pool 1, IMS/VoNR, OSS, lawful intercept; interconnect and roaming tests; first call M4
C2	M6–M9	Geo-redundancy failover tests DC1↔DC2; Amman Centre and Zarqa edge UPF + CDN (with RO-02)
C3	M8–M12	Irbid regional DC: edge UPF/MEC, UDR replica (with RO-02/03); Aqaba regional DC and international gateway (with RO-03)
C4	M13–M30	UPF capacity steps following traffic: +400 Gbps per 600 accepted city sites; slice catalogue (eMBB-1G GFBR, FWA, enterprise); NEF exposure
C5	M31–M36	Final dimensioning to $\approx 2,492$ Gbps busy hour (N+1), DR drill, hand-over

2. Backbone and metro core (must be lit before the dependent rollout opens)

Rollout	Routes lit	Route km
RO-01	Amman metro core ring (87 km)	87
RO-02	Amman-Zarqa-Mafraq-Jaber (74 km); Mafraq-Ramtha-Irbid (52 km); Irbid-Jerash-Amman (R35) (68 km)	194
RO-03	Desert Highway (R15) Amman-Aqaba (340 km)	340
RO-04	Dead Sea - Wadi Araba (R65) Amman-Aqaba (337 km)	337
RO-05	King's Highway Madaba-Karak-Tafilah-Petra-Ma'an (241 km); Amman-Salt-Deir Alla (37 km)	278
RO-06	Jordan Valley (R65 north) (127 km); Irbid-Ajloun-Jerash (47 km)	174
RO-07	Karak-Qatraneh (39 km); Tafilah-Jurf (34 km); Amman-Azraq (R40) (112 km)	185
RO-08	Zarqa-Azraq-Safawi (137 km); Mafraq-Safawi-Ruwaished (R10) (302 km); Quweira-Wadi Rum (32 km)	471
RO-09	–	0
RO-10	–	0

Sequence logic: metro core (RO-01) → northern ring Amman–Zarqa–Mafraq–Irbid–Jerash (RO-02) → Desert Highway to Aqaba incl. cable-landing access (RO-03) → Wadi Araba–Dead Sea closes the southern national ring (RO-04) → King's Highway and Salt (RO-05) → Jordan Valley, Ajloun (RO-06) → cross-links and Azraq (RO-07) → eastern desert and Wadi Rum (RO-08). First light may use leased/swapped dark fiber; DWDM/OTN nodes at every PoP; ring protection verified (< 50 ms) before the rollout gate G2.

3. Hubs, aggregation and access rings

Rollout	Hubs commissioned	Hub routers	Access-ring km	Aggregation km	Spur km	Microwave hops	Pooled DUs (C-RAN)	Site DUs (D-RAN)
RO-01	6	12	228	32	0	0	230	0
RO-02	10	20	381	64	0	0	349	0
RO-03	15	30	537	108	0	0	453	14
RO-04	16	32	611	108	30	0	515	33
RO-05	19	38	697	109	30	0	513	85

Rollout	Hubs commissioned	Hub routers	Access-ring km	Aggregation km	Spur km	Microwave hops	Pooled DUs (C-RAN)	Site DUs (D-RAN)
RO-06	19	38	810	106	6	0	448	173
RO-07	13	26	463	53	60	246	320	340
RO-08	18	36	667	107	28	107	99	586
RO-09	17	34	764	121	47	7	0	669
RO-10	17	34	579	80	0	0	0	471
Total	150	300	5,737	888	201	360	2927	2,371

Build order inside a rollout: (1) hub room (power 2N, cooling, GNSS, racks) → (2) aggregation ring to PoP, routers, PTP grandmaster/boundary-clock chain verified ($\pm 1.5 \mu\text{s}$; $\pm 130 \text{ ns}$ inside C-RAN cluster) → (3) access rings, both directions spliced and OTDR-tested before radio integration → (4) DU pool commissioning → (5) sites. A ring is accepted only when closed: single-homed sites are not counted as on-air for gate G3.

Fiber acceptance: OTDR both directions 1310/1550 nm, splice $\leq 0.1 \text{ dB}$, span loss within budget for 25GE-LR / eCPRI ($\leq 10 \text{ km}$ fronthaul), as-built GIS route uploaded.

03 – Site Implementation Standard (survey → acceptance)

1. Process and target durations

Step	Output	Target	Owner
1. Nominal release	Search ring (0.2 × ISD), target height, azimuth/tilt from output/sectors.csv	T-6	RF planning
2. Candidate search, TSSR	≥ 2 candidates; TSSR with panoramic photos, roof height, structural notes, line-of-sight to neighbours, power and fiber entry	3 weeks	Acquisition + RF + civil
3. RF validation	Candidate accepted if height ≥ nominal -3 m, offset ≤ search ring, no main-beam blockage within 100 m; re-run prediction if moved > 0.15 × ISD	3 days	RF planning
4. Lease and permits	Lease, municipality permit, EMF file, civil-aviation clearance near QAIA/Marka/Aqaba airports	6–10 weeks (shared site 2–4)	Acquisition
5. Detailed design	Structural calculation (rooftop loading for 64T64R AAU + passive antenna), power, grounding, fiber entry drawings	2 weeks	Civil
6. Civil and power	Poles/monopole/tower, cabinets, rectifier + Li-ion backup (4 h city, 8 h rural + solar hybrid off-grid), grid connection	2–6 weeks	Civil contractor
7. Fiber / microwave	Drop cable to ring, splice, OTDR; or microwave LOS survey, install, link test	parallel with 6	Transport
8. Radio install	AAU, passive antenna, RRUs, DU or fronthaul, GNSS; azimuth ±3°, mechanical tilt ±0.5° with digital inclinometer/AAT photos	2–3 days	RAN vendor
9. Commissioning and integration	Software, licences, cell parameters (PCI, RSI, TAC, neighbours, RET, digital tilt), alarms clear, integration to 5GC and OSS	1 day	RAN vendor
10. SSV	Single-site verification (section 5)	≤ 5 days after on-air	RAN vendor
11. Cluster optimisation and acceptance	Section 6	when ≥ 90 % of cluster on air	Optimisation
12. Hand-over	As-built pack, asset register, spares, O&M acceptance	T+4	PMO

2. Configuration by site type

Type	Structure	Radio line-up per sector
City rooftop	3–6 m poles or stub tower so antenna ≥ roof max + 8 m	n78 64T64R AAU + 2L4H passive antenna, n28 RRU, n1/n3 dual-band RRU
City monopole	25–35 m monopole, camouflaged where required by municipality	same
Rural tower	45 m lattice	2.6 m passive antenna, n28 + n1/n3 4T4R, n78 8T8R
Highway tower	45 m lattice, 2 sectors along road	n28 + n3 4T4R

3. Installation quality checklist (photo evidence mandatory)

Antenna azimuth, mechanical tilt, height label · RET calibrated and addressable · AAU fan clearance and bracket torque · connector torque and weatherproofing · grounding ≤ 5 Ω, lightning protection · fiber bend radius and

labelling both ends · GNSS sky view · cabinet sealing, battery test · site tidy, EMF signage, access safety (roof edge protection, ladder cage).

4. HSE

Work-at-height certification for all riggers, permit-to-work per site, RF-off procedure on shared structures, daily toolbox talk, heat-stress rules (no tower climbing 12:00–15:00 June–August), traffic management for street fiber works, zero-fatality target, LTIFR < 0.5.

5. Single-site verification (per sector unless stated)

Test	Pass criterion
Alarms / VSWR / RET / GNSS lock	none active / < 1.4 / responds / locked
Sector swap and azimuth check (drive around)	PCI footprint matches plan
Static DL near-cell (SS-RSRP $\geq$ -80 dBm, 3CC n78 + FDD CA)	$\geq$ 1.5 Gbps peak, $\geq$ 1.2 Gbps 30 s average
Static UL	$\geq$ 100 Mbps
Latency to edge UPF	$\leq$ 15 ms RTT
VoNR call set-up, 2 min hold	success, MOS $\geq$ 4.0
Intra-site and inter-site handover	100 % success on test route
Backhaul	25GE link up, PTP locked, throughput test $\geq$ 8 Gbps

6. Cluster acceptance – the 1 Gbps KPI

Drive test on all accessible roads of the cluster with a 3CC-capable UE, network in live low-load condition (night or pre-launch), 50 m bins: | KPI | Target | |---|---| | Bins with single-user DL $\geq$ 1 Gbps | $\geq$ **95 %** dense urban / urban, $\geq$ 93 % suburban | | 5th-percentile DL | $\geq$ 700 Mbps | | SS-RSRP $\geq$ -105 dBm | $\geq$ 98 % of bins | | SS-SINR $\geq$ 13 dB | $\geq$ 90 % of bins | | UL $\geq$ 20 Mbps | $\geq$ 95 % of bins | | Session set-up success / drop rate | $\geq$ 99.5 % / $\leq$ 0.3 % | | Handover success | $\geq$ 99.5 % | | Stationary indoor sample (10 light-indoor points per cluster) | $\geq$ 80 % of points $\geq$ 1 Gbps | Failing bins are analysed against the terrain-aware prediction (map layer “1 Gbps compliance”): actions in order – digital tilt/beam-set change, azimuth change, neighbour/PCI fix, add n258 or infill site (change request). Rural/highway acceptance: n28 SS-RSRP $\geq$ -110 dBm on $\geq$ 95 % of the route, DL $\geq$ 10 Mbps, no call drop on the corridor.

After launch the operating KPI is busy-hour PRB utilisation: > 20 % on a sector for 4 weeks triggers capacity action, protecting the single-user 1 Gbps experience.

04 – Indoor Deployment Plan

51 venues · 5,339 n78 pRRUs · 196 n258 heads. A venue is scheduled one rollout after the macro cluster around it (earliest RO-02) so neighbours, handover and leakage control can be tuned against a live outdoor layer.

Rollout	Venues	pRRU	n258 heads
RO-01	0	0	0
RO-02	18	1740	58
RO-03	5	573	38
RO-04	14	1545	47
RO-05	1	88	4
RO-06	9	650	28
RO-07	0	0	0
RO-08	2	319	4
RO-09	0	0	0
RO-10	2	424	17

Venue schedule

Rollout	Window	Venue	Type	n78 pRRU	n258 heads
RO-02	Jul 2027 – Sep 2027	Abdali Mall	mall	248	10
RO-02	Jul 2027 – Sep 2027	The Boulevard - Abdali	mixed-use	110	4
RO-02	Jul 2027 – Sep 2027	Galleria Mall	mall	66	4
RO-02	Jul 2027 – Sep 2027	Al Baraka Mall	mall	50	4
RO-02	Jul 2027 – Sep 2027	Istiklal Mall	mall	66	4
RO-02	Jul 2027 – Sep 2027	Four Seasons Amman	hotel	66	2
RO-02	Jul 2027 – Sep 2027	The St. Regis Amman	hotel	88	2
RO-02	Jul 2027 – Sep 2027	The Ritz-Carlton Amman	hotel	79	2
RO-02	Jul 2027 – Sep 2027	Fairmont Amman	hotel	103	4
RO-02	Jul 2027 – Sep 2027	W Amman	hotel	86	2
RO-02	Jul 2027 – Sep 2027	Amman Rotana	hotel	125	4
RO-02	Jul 2027 – Sep 2027	InterContinental Jordan	hotel	130	4
RO-02	Jul 2027 – Sep 2027	Amman Marriott	hotel	89	2
RO-02	Jul 2027 – Sep 2027	Kempinski Amman	hotel	83	2
RO-02	Jul 2027 – Sep 2027	Le Royal Amman	hotel	105	2
RO-02	Jul 2027 – Sep 2027	Sheraton Amman	hotel	84	2
RO-02	Jul 2027 – Sep 2027	Landmark Amman	hotel	79	2
RO-02	Jul 2027 – Sep 2027	Crowne Plaza Amman	hotel	83	2
RO-03	Oct 2027 – Dec 2027	Taj Lifestyle Center	mall	164	6
RO-03	Oct 2027 – Dec 2027	Grand Hyatt Amman	hotel	99	4
RO-03	Oct 2027 – Dec 2027	Al-Bashir Hospital	hospital	130	2
RO-03	Oct 2027 – Dec 2027	Amman Intl Stadium / Al-Hussein Youth City	stadium	50	24
RO-03	Oct 2027 – Dec 2027	Yarmouk University	campus	130	2

Rollout	Window	Venue	Type	n78 pRRU	n258 heads
RO-04	Jan 2028 – Mar 2028	City Mall	mall	175	7
RO-04	Jan 2028 – Mar 2028	Mecca Mall	mall	208	8
RO-04	Jan 2028 – Mar 2028	King Hussein Business Park	campus	130	2
RO-04	Jan 2028 – Mar 2028	University of Jordan	campus	270	2
RO-04	Jan 2028 – Mar 2028	Jordan University Hospital	hospital	103	2
RO-04	Jan 2028 – Mar 2028	King Hussein Cancer Center	hospital	119	2
RO-04	Jan 2028 – Mar 2028	Ayla Oasis / Hyatt Regency	resort	112	2
RO-04	Jan 2028 – Mar 2028	Al Manara - Saraya Aqaba	hotel	69	2
RO-04	Jan 2028 – Mar 2028	Kempinski Aqaba	hotel	64	2
RO-04	Jan 2028 – Mar 2028	InterContinental Aqaba	hotel	81	2
RO-04	Jan 2028 – Mar 2028	Movenpick Aqaba	hotel	90	2
RO-04	Jan 2028 – Mar 2028	Aqaba Gateway / City Center	mall	44	4
RO-04	Jan 2028 – Mar 2028	King Hussein Intl Airport (Aqaba)	airport	25	6
RO-04	Jan 2028 – Mar 2028	Arabella Mall Irbid	mall	55	4
RO-05	Apr 2028 – Jun 2028	Irbid City Centre	mall	88	4
RO-06	Jul 2028 – Sep 2028	Kempinski Ishtar Dead Sea	hotel	111	4
RO-06	Jul 2028 – Sep 2028	Movenpick Dead Sea	hotel	109	4
RO-06	Jul 2028 – Sep 2028	Dead Sea Marriott	hotel	80	2
RO-06	Jul 2028 – Sep 2028	Hilton Dead Sea	hotel	90	2
RO-06	Jul 2028 – Sep 2028	Crowne Plaza Dead Sea	hotel	126	4
RO-06	Jul 2028 – Sep 2028	King Hussein Bin Talal Convention Centre	convention	42	6
RO-06	Jul 2028 – Sep 2028	Movenpick Petra	hotel	52	2
RO-06	Jul 2028 – Sep 2028	Petra Marriott	hotel	31	2
RO-06	Jul 2028 – Sep 2028	Petra Visitor Centre	venue	9	2
RO-08	Jan 2029 – Mar 2029	JUST campus	campus	216	2
RO-08	Jan 2029 – Mar 2029	King Abdullah University Hospital	hospital	103	2
RO-10	Jul 2029 – Sep 2029	Queen Alia Intl Airport terminal	airport	172	13
RO-10	Jul 2029 – Sep 2029	Tala Bay resort cluster	resort	252	4

Per-venue process (14–18 weeks)

1. Access agreement with the owner (commercial model: neutral-host ready, operator-funded) – 4–6 weeks, started at T–6.
2. Walk survey, floor plans, 3-D indoor prediction, pRRU/RHUB positions, cable routes; owner design approval – 3 weeks.
3. Installation out of trading hours: CAT6A/hybrid fiber, RHUBs in IDF rooms, DU in MDF, dual diverse fiber to hub – 4–6 weeks.
4. Commissioning, walk test: $\geq 95\%$ of public area ≥ 1 Gbps, SS-RSRP ≥ -95 dBm, leakage outside ≤ -105 dBm at 10 m or 10 dB below macro, lifts and car parks VoNR continuity – 1–2 weeks.
5. Hand-over with as-built drawings and owner O&M contact sheet. Phase 2 (after RO-10, outside this plan's BoQ): every building $> 10,000$ m² or > 8 floors identified from the building-height layer.

05 – Resource and Procurement Plan

1. Field resources per rollout

Rollout	Sites	Acquisition agents	Civil crews	Urban fiber crews	Backbone crews	Install crews	SSV teams	Optimisation teams
RO-01	306	9	51	22	1	15	2	3
RO-02	465	13	80	38	2	23	3	4
RO-03	617	18	125	54	4	30	4	5
RO-04	719	20	181	63	4	35	4	5
RO-05	769	22	202	70	3	37	5	6
RO-06	770	22	221	77	2	37	5	6
RO-07	766	22	278	48	2	37	5	6
RO-08	718	20	229	67	5	35	4	5
RO-09	669	19	218	78	0	32	4	5
RO-10	471	14	150	55	0	23	3	4

Productivity norms: acquisition agent 12 sites / 3 months; civil crew 2 rooftops, 1 monopole or 0.7 tower per month; urban fiber crew 4 route-km/month (micro-trench / existing duct); backbone crew 25 km/month (plough / duct along highways); install crew 7 sites/month; SSV team 60 sites/month; one optimisation team per 150 sites. Three to four rollouts overlap, so **programme peak $\approx 1.6 \times$ the single-rollout peak** for acquisition and civil, $1.2 \times$ for install. Minimum three civil and two fiber contractors to avoid single-supplier dependency; regional sub-contractors for Irbid, Aqaba and the south.

2. Bill of quantities by rollout

item	RO-01	RO-02	RO-03	RO-04	RO-05	RO-06	RO-07	RO-08	RO-09	RO-10	total
Macro sites	306	465	617	719	769	770	766	718	669	471	6270
– rooftop	306	452	487	385	359	225	132	181	83	42	2652
– monopole / greenfield	0	13	130	301	378	538	323	400	528	429	3040
– lattice tower 45 m	0	0	0	33	32	7	311	137	58	0	578
Sectors	918	1395	1851	2124	2275	2303	2298	2142	1956	1413	18675
n78 64T64R AAU (3×100 MHz)	918	1395	1851	2058	2211	2289	1365	1743	1833	1413	17076
n78 8T8R radio + antenna (rural)	0	0	0	0	0	0	933	377	21	0	1331
Passive multiband antenna 2L4H + RET	918	1395	1851	2124	2275	2303	2298	2142	1956	1413	18675
n28 RRU	918	1395	1851	2124	2275	2303	2298	2142	1956	1413	18675
n1/n3 dual-band RRU	918	1395	1851	2124	2275	2303	2298	2142	1956	1413	18675
n258 street small cells	0	0	0	0	0	0	0	0	480	480	960
DU (pooled, C-RAN hub)	230	349	453	515	513	448	320	99	0	0	2927
DU (at site, D-RAN)	0	0	14	33	85	173	340	586	669	471	2371
Cell-site router 25GE (D-RAN sites)	0	0	14	33	85	173	340	586	669	471	2371
Fronthaul gateway / passive WDM (C-RAN sites)	306	465	603	686	684	597	426	132	0	0	3899
Hubs (DU hotel / pre-agg) commissioned	6	10	15	16	19	19	13	18	17	17	150

item	RO-01	RO-02	RO-03	RO-04	RO-05	RO-06	RO-07	RO-08	RO-09	RO-10	total
Hub aggregation routers 100GE	12	20	30	32	38	38	26	36	34	34	300
Microwave hops	0	0	0	0	0	0	246	107	7	0	360
Access-ring fiber (route km)	228	381	537	611	697	810	463	667	764	579	5737
Aggregation-ring fiber (route km)	32	64	108	108	109	106	53	107	121	80	888
Backbone / metro-core fiber lit (route km)	87	194	340	337	278	174	185	471	0	0	2066
Backbone add/drop spurs (km)	0	0	0	30	30	6	60	28	47	0	201
Indoor venues	0	18	5	14	1	9	0	2	0	2	51
Indoor n78 pRRU	0	1740	573	1545	88	650	0	319	0	424	5339
Indoor n258 heads	0	58	38	47	4	28	0	4	0	17	196

3. Lead times and call-off calendar

Item	Lead time	Order rule
64T64R AAU, RRUs, DU/BBU	16–20 weeks	frame order M1; firm call-off per rollout at gate G1 (T–3) with 10 % buffer; RO-01...RO-03 called off together at M1
Passive antennas, RET, jumpers	10–12 weeks	same as radio
Routers, DWDM/OTN	16–24 weeks	backbone nodes ordered one rollout earlier than radio
Fiber cable (48/96-core), ducts	8–12 weeks	annual volume order, monthly deliveries
Monopoles / towers (local steel)	8–10 weeks	rolling 3-month forecast to two local fabricators
Rectifiers, Li-ion batteries, solar hybrid	12–16 weeks	with radio
pRRU / RHUB / mmWave heads	16 weeks	per venue design approval
Core / cloud hardware	12–16 weeks	M1 (C1), then per capacity step

4. Logistics

Central warehouse Amman (Sahab / Al-Muwaqqar industrial area) from M2; regional cross-docks Irbid (RO-02) and Aqaba (RO-03, also port of entry with ASEZA customs regime). Kitting per site (one pallet set per site, barcoded to the site ID in `rollout_sites.csv`); reverse logistics for packaging and faulty units; 2 % spares pool growing with the installed base.

5. People and competence

PMO ≈ 45 staff at peak. Vendor training academy in M1–M3: 5G SA commissioning, massive-MIMO beam configuration, eCPRI/PTP, work-at-height. Local content target ≥ 70 % of field hours.

06 – Risk Register

P = probability, I = impact (1 low – 5 high).

#	Risk	P	I	Rollouts	Mitigation	Owner
1	Less than 300 MHz of n78 awarded	3	5	all	Design fallback: 200 MHz gives $\approx 78\%$ of bins ≥ 1 Gbps – bring the n258 layer forward from RO-09 to RO-03 in dense urban, enable DL CoMP in C-RAN clusters; adjust commercial promise by area	Regulatory / CTO
2	Site acquisition slower than 3 months (rooftop owners, municipality permits)	4	4	RO-01...RO-06	Sharing frameworks first, 2 candidates per nominal, GAM/ASEZA fast-track MoU, standard lease and rent card, acquisition starts T-6	Acquisition
3	Hilltop/rooftop structural limits for 64T64R + passive antenna (≈ 75 kg per sector)	3	3	city rollouts	Structural screening in TSSR, lightweight poles, split to two roofs, 32T32R variant where loading fails	Civil
4	Fiber right-of-way and street-works permits delay rings	4	4	all city rollouts	Use existing ducts / lease dark fiber first, micro-trenching approval, ring may open on E-band protection for ≤ 60 days	Transport
5	Backbone route not lit before dependent rollout	3	4	RO-02...RO-05, RO-08	Leased capacity as first light; backbone ordered one rollout ahead; gate G2 checks	Transport
6	Supply-chain delay of AAUs / routers	3	4	all	Frame agreement with delivery SLA and penalties, three rollouts of buffer stock in year 1, dual-source routers	Procurement
7	Grid connection lead time; rural sites off-grid	3	3	RO-04...RO-09	Utility framework, temporary generator ≤ 60 days, solar-hybrid standard design for rural/highway	Civil
8	Cross-border interference / coordination (109 sectors) and TDD frame misalignment with neighbours	3	3	RO-03...RO-06 (Aqaba, Jordan Valley, Ramtha)	Early TRC coordination, $+2^\circ$ tilt and power limits already in design, common DDDSU frame, monitoring probes	Regulatory / RF
9	1 Gbps cluster acceptance fails (model vs reality in hilly, stone-built Amman)	3	4	RO-01...RO-03	CW calibration campaign in M1-M3, RO-01 used as pilot with 3-week optimisation buffer, infill budget 3 % of sites	RF
10	Device ecosystem: few UEs support 3CC n78 CA	3	4	launch	CPE-first launch (FWA), device certification programme with OEMs, 2CC fallback marketed by area	Marketing
11	Core readiness slips (M4 first call)	2	5	RO-01	Vendor-hosted staging core from M2, phased NF activation, parallel DC2 build	Core
12	HSE incident at height or in street works	2	5	all	Certification, permit-to-work, audits 10 % of sites weekly, stop-work authority	HSE
13	Community / EMF objections	3	2	city rollouts	Public EMF measurements portal, camouflage catalogue, municipality engagement	Corporate affairs
14	Resource peak (≈ 37 install + 278 civil crews) not available locally	3	3	RO-04...RO-08	Training academy M1-M3, regional contractors, levelled quotas (≤ 770 sites per rollout)	PMO
15	Currency / budget overrun from survey-driven scope change ($\pm 10\%$ sites)	3	3	all	Re-baseline at each G1, contingency 8 %, change control board	PMO / Finance
16	Indoor venue owners delay access	3	2	RO-02...RO-10	Start agreements at T-6, neutral-host offer, substitute venues from phase-2 list	Indoor

RO-01 – Work Package

On-air window: Apr 2027 – Jun 2027 (M4–M6) · 306 sites · 918 sectors · 6 hubs · 260 km access/aggregation fiber · 87 km backbone · 0 indoor venues

After this rollout: 306 sites on air (5 % of programme), ≈ 8.6 % of population covered, 40 km² of 1 Gbps footprint.
Site list: filter `rollout = RO-01` in `rollout_sites.csv` / workbook sheet “Sites”; on the map choose “Rollout phase” and move the slider to 1.

1. Scope

Class	Sites	Site type	Sites
Dense urban	293	Rooftop	306
Urban	13	Monopole / greenfield	0
Suburban	0	Lattice tower	0
Rural	0		
Highway	0	Total	306

Areas: Amman-Zarqa Metro (306). **Governorates:** Amman 306.

2. Work packages (build units)

Unit	Type	Cluster / route	Sites
HUB-AMM-28	Hub cluster	Amman-Zarqa Metro	61
HUB-AMM-69	Hub cluster	Amman-Zarqa Metro	55
HUB-AMM-14	Hub cluster	Amman-Zarqa Metro	51
HUB-AMM-82	Hub cluster	Amman-Zarqa Metro	48
HUB-AMM-13	Hub cluster	Amman-Zarqa Metro	47
HUB-AMM-38	Hub cluster	Amman-Zarqa Metro	44

3. Bill of quantities

Item	Qty	Item	Qty
n78 64T64R AAU	918	Hubs / DU hotels	6
n78 8T8R (rural)	0	Hub aggregation routers	12
Passive 2L4H antenna + RET	918	Cell-site routers 25GE	0
n28 RRU	918	Access-ring fiber km	228
n1/n3 RRU	918	Aggregation fiber km	32
Pooled DU (C-RAN, 0.75 per site)	230	Backbone add/drop spur km	0
Site DU (D-RAN)	0	Backbone / metro core km lit	87
Fronthaul gateways (C-RAN)	306		
n258 street cells	0	Microwave hops	0
Indoor pRRU / n258 heads	0 / 0	Power systems (rectifier + battery)	306

4. Prerequisites (entry criteria)

- Spectrum licence and type approval
- DC1/DC2 core ready, first call (M4)
- Amman metro core ring lit
- Site-sharing and dark-fiber frameworks signed
- CW model-calibration campaign complete
- Backbone lit in this rollout before G2: Amman metro core ring (87 km)

5. Schedule

Activity	Start	End
Nominal release, TSSR surveys, site acquisition & permits	M1 · Jan 2027	M1 · Jan 2027
Detailed RF / transport design freeze, material call-off	M1 · Jan 2027	M1 · Jan 2027
Civil works, power, fiber build to hubs and rings	M1 · Jan 2027	M3 · Mar 2027
Hub / DU-hotel and router commissioning	M2 · Feb 2027	M4 · Apr 2027
Radio install, commissioning, integration to 5GC	M3 · Mar 2027	M6 · Jun 2027
Single-site verification (SSV)	M4 · Apr 2027	M6 · Jun 2027
Cluster optimisation & acceptance (1 Gbps drive test)	M5 · May 2027	M8 · Aug 2027
Commercial launch of rollout area / hand-over to operations	M7 · Jul 2027	M8 · Aug 2027

Required run-rate: $\approx$ 102 sites on air per month; $\approx$ 87 km of access/aggregation fiber per month.

6. Resources

9 acquisition agents · 51 civil crews · 22 urban fiber crews · 1 backbone crews · 15 radio install crews · 2 SSV teams · 3 optimisation teams.

7. Indoor venues in this rollout

None.

8. Exit criteria

- G3: $\geq$ 95 % of sites on air, all access rings closed, no open critical alarms.
- G4: every cluster passes the acceptance KPIs of 03_Site_Implementation_Standard.md §6 ($\geq$ 95 % of bins $\geq$ 1 Gbps in DU/U, $\geq$ 93 % SU; rural/highway route KPIs).
- G5: as-built pack, asset register, spares and O&M hand-over signed; lessons-learned fed to RO-02.

9. Rollout-specific notes

- Dense-urban rooftops: structural screening and owner negotiations dominate; C-RAN fronthaul $\leq$ 10 km to the DU hotel must be verified per site.
- More than half the sites are rooftops – prioritise existing shared rooftops; antenna must clear local roofs by 8 m (heights in rollout_sites.csv).
- Pilot rollout: 3-week extra optimisation buffer; acceptance results recalibrate the prediction model for all later rollouts.

RO-02 – Work Package

On-air window: Jul 2027 – Sep 2027 (M7–M9) · 465 sites · 1395 sectors · 10 hubs · 445 km access/aggregation fiber · 194 km backbone · 18 indoor venues

After this rollout: 771 sites on air (12 % of programme), ≈ 18.4 % of population covered, 117 km² of 1 Gbps footprint. Site list: filter `rollout = RO-02` in `rollout_sites.csv` / workbook sheet “Sites”; on the map choose “Rollout phase” and move the slider to 2.

1. Scope

Class	Sites	Site type	Sites
Dense urban	176	Rooftop	452
Urban	289	Monopole / greenfield	13
Suburban	0	Lattice tower	0
Rural	0		
Highway	0	Total	465

Areas: Amman-Zarqa Metro (375), Irbid (90). **Governorates:** Amman 375, Irbid 90.

2. Work packages (build units)

Unit	Type	Cluster / route	Sites
HUB-AMM-52	Hub cluster	Amman-Zarqa Metro	54
HUB-IRB-15	Hub cluster	Irbid	52
HUB-AMM-57	Hub cluster	Amman-Zarqa Metro	51
HUB-AMM-41	Hub cluster	Amman-Zarqa Metro	50
HUB-AMM-45	Hub cluster	Amman-Zarqa Metro	46
HUB-AMM-59	Hub cluster	Amman-Zarqa Metro	45
HUB-AMM-23	Hub cluster	Amman-Zarqa Metro	43
HUB-AMM-64	Hub cluster	Amman-Zarqa Metro	43
HUB-AMM-35	Hub cluster	Amman-Zarqa Metro	43
HUB-IRB-01	Hub cluster	Irbid	38

3. Bill of quantities

Item	Qty	Item	Qty
n78 64T64R AAU	1395	Hubs / DU hotels	10
n78 8T8R (rural)	0	Hub aggregation routers	20
Passive 2L4H antenna + RET	1395	Cell-site routers 25GE	0
n28 RRU	1395	Access-ring fiber km	381
n1/n3 RRU	1395	Aggregation fiber km	64
Pooled DU (C-RAN, 0.75 per site)	349	Backbone add/drop spur km	0
Site DU (D-RAN)	0	Backbone / metro core km lit	194
Fronthaul gateways (C-RAN)	465		

Item	Qty	Item	Qty
n258 street cells	0	Microwave hops	0
Indoor pRRU / n258 heads	1740 / 58	Power systems (rectifier + battery)	465

4. Prerequisites (entry criteria)

- RO-01 gate G2 passed (teams and process release)
- Backbone lit in this rollout before G2: Amman-Zarqa-Mafraq-Jaber (74 km); Mafraq-Ramtha-Irbid (52 km); Irbid-Jerash-Amman (R35) (68 km)
- Backbone already in service from earlier rollouts: Amman metro core ring

5. Schedule

Activity	Start	End
Nominal release, TSSR surveys, site acquisition & permits	M1 · Jan 2027	M4 · Apr 2027
Detailed RF / transport design freeze, material call-off	M2 · Feb 2027	M4 · Apr 2027
Civil works, power, fiber build to hubs and rings	M3 · Mar 2027	M6 · Jun 2027
Hub / DU-hotel and router commissioning	M5 · May 2027	M7 · Jul 2027
Radio install, commissioning, integration to 5GC	M6 · Jun 2027	M9 · Sep 2027
Single-site verification (SSV)	M7 · Jul 2027	M9 · Sep 2027
Cluster optimisation & acceptance (1 Gbps drive test)	M8 · Aug 2027	M11 · Nov 2027
Commercial launch of rollout area / hand-over to operations	M10 · Oct 2027	M11 · Nov 2027

Required run-rate: $\approx$ 155 sites on air per month; $\approx$ 149 km of access/aggregation fiber per month.

6. Resources

13 acquisition agents · 80 civil crews · 38 urban fiber crews · 2 backbone crews · 23 radio install crews · 3 SSV teams · 4 optimisation teams.

7. Indoor venues in this rollout

Venue	Type	pRRU	n258
Abdali Mall	mall	248	10
The Boulevard - Abdali	mixed-use	110	4
Galleria Mall	mall	66	4
Al Baraka Mall	mall	50	4
Istiklal Mall	mall	66	4
Four Seasons Amman	hotel	66	2
The St. Regis Amman	hotel	88	2
The Ritz-Carlton Amman	hotel	79	2
Fairmont Amman	hotel	103	4
W Amman	hotel	86	2
Amman Rotana	hotel	125	4
InterContinental Jordan	hotel	130	4
Amman Marriott	hotel	89	2

Venue	Type	pRRU	n258
Kempinski Amman	hotel	83	2
Le Royal Amman	hotel	105	2
Sheraton Amman	hotel	84	2
Landmark Amman	hotel	79	2
Crowne Plaza Amman	hotel	83	2

8. Exit criteria

- G3: $\geq 95\%$ of sites on air, all access rings closed, no open critical alarms.
- G4: every cluster passes the acceptance KPIs of 03_Site_Implementation_Standard.md §6 ($\geq 95\%$ of bins ≥ 1 Gbps in DU/U, $\geq 93\%$ SU; rural/highway route KPIs).
- G5: as-built pack, asset register, spares and O&M hand-over signed; lessons-learned fed to RO-03.

9. Rollout-specific notes

- Dense-urban rooftops: structural screening and owner negotiations dominate; C-RAN fronthaul ≤ 10 km to the DU hotel must be verified per site.
- More than half the sites are rooftops – prioritise existing shared rooftops; antenna must clear local roofs by 8 m (heights in rollout_sites.csv).

RO-03 – Work Package

On-air window: Oct 2027 – Dec 2027 (M10–M12) · 617 sites · 1851 sectors · 15 hubs · 645 km access/aggregation fiber · 340 km backbone · 5 indoor venues

After this rollout: 1,388 sites on air (22 % of programme), $\approx$ 28.5 % of population covered, 236 km² of 1 Gbps footprint. Site list: filter `rollout = RO-03` in `rollout_sites.csv` / workbook sheet “Sites”; on the map choose “Rollout phase” and move the slider to 3.

1. Scope

Class	Sites	Site type	Sites
Dense urban	8	Rooftop	487
Urban	589	Monopole / greenfield	130
Suburban	20	Lattice tower	0
Rural	0		
Highway	0	Total	617

Areas: Amman-Zarqa Metro (456), Aqaba (123), Irbid (38). **Governorates:** Amman 444, Aqaba 123, Irbid 38, Balqa 12.

2. Work packages (build units)

Unit	Type	Cluster / route	Sites
HUB-AQB-05	Hub cluster	Aqaba	57
HUB-AQB-06	Hub cluster	Aqaba	52
HUB-AMM-56	Hub cluster	Amman-Zarqa Metro	49
HUB-AMM-83	Hub cluster	Amman-Zarqa Metro	48
HUB-AMM-47	Hub cluster	Amman-Zarqa Metro	48
HUB-AMM-44	Hub cluster	Amman-Zarqa Metro	44
HUB-AMM-79	Hub cluster	Amman-Zarqa Metro	41
HUB-AMM-05	Hub cluster	Amman-Zarqa Metro	40
HUB-AMM-02	Hub cluster	Amman-Zarqa Metro	39
HUB-IRB-05	Hub cluster	Irbid	38
HUB-AMM-49	Hub cluster	Amman-Zarqa Metro	38
HUB-AMM-11	Hub cluster	Amman-Zarqa Metro	37
HUB-AMM-27	Hub cluster	Amman-Zarqa Metro	36
HUB-AMM-04	Hub cluster	Amman-Zarqa Metro	36
HUB-AQB-07	Hub cluster	Aqaba	14

3. Bill of quantities

Item	Qty	Item	Qty
n78 64T64R AAU	1851	Hubs / DU hotels	15
n78 8T8R (rural)	0	Hub aggregation routers	30
Passive 2L4H antenna + RET	1851	Cell-site routers 25GE	14

Item	Qty	Item	Qty
n28 RRU	1851	Access-ring fiber km	537
n1/n3 RRU	1851	Aggregation fiber km	108
Pooled DU (C-RAN, 0.75 per site)	453	Backbone add/drop spur km	0
Site DU (D-RAN)	14	Backbone / metro core km lit	340
Fronthaul gateways (C-RAN)	603		
n258 street cells	0	Microwave hops	0
Indoor pRRU / n258 heads	573 / 38	Power systems (rectifier + battery)	617

4. Prerequisites (entry criteria)

- RO-02 gate G2 passed (teams and process release)
- Backbone lit in this rollout before G2: Desert Highway (R15) Amman-Aqaba (340 km)
- Backbone already in service from earlier rollouts: Amman metro core ring; Amman-Zarqa-Mafraq-Jaber; Irbid-Jerash-Amman; Mafraq-Ramtha-Irbid
- Cross-border coordination filed for 64 flagged sectors in this rollout

5. Schedule

Activity	Start	End
Nominal release, TSSR surveys, site acquisition & permits	M4 · Apr 2027	M7 · Jul 2027
Detailed RF / transport design freeze, material call-off	M5 · May 2027	M7 · Jul 2027
Civil works, power, fiber build to hubs and rings	M6 · Jun 2027	M9 · Sep 2027
Hub / DU-hotel and router commissioning	M8 · Aug 2027	M10 · Oct 2027
Radio install, commissioning, integration to 5GC	M9 · Sep 2027	M12 · Dec 2027
Single-site verification (SSV)	M10 · Oct 2027	M12 · Dec 2027
Cluster optimisation & acceptance (1 Gbps drive test)	M11 · Nov 2027	M14 · Feb 2028
Commercial launch of rollout area / hand-over to operations	M13 · Jan 2028	M14 · Feb 2028

Required run-rate: $\approx$ 206 sites on air per month; $\approx$ 215 km of access/aggregation fiber per month.

6. Resources

18 acquisition agents · 125 civil crews · 54 urban fiber crews · 4 backbone crews · 30 radio install crews · 4 SSV teams · 5 optimisation teams.

7. Indoor venues in this rollout

Venue	Type	pRRU	n258
Taj Lifestyle Center	mall	164	6
Grand Hyatt Amman	hotel	99	4
Al-Bashir Hospital	hospital	130	2
Amman Intl Stadium / Al-Hussein Youth City	stadium	50	24
Yarmouk University	campus	130	2

8. Exit criteria

- G3: $\geq 95\%$ of sites on air, all access rings closed, no open critical alarms.
- G4: every cluster passes the acceptance KPIs of 03_Site_Implementation_Standard.md §6 ($\geq 95\%$ of bins ≥ 1 Gbps in DU/U, $\geq 93\%$ SU; rural/highway route KPIs).
- G5: as-built pack, asset register, spares and O&M hand-over signed; lessons-learned fed to RO-04.

9. Rollout-specific notes

- More than half the sites are rooftops – prioritise existing shared rooftops; antenna must clear local roofs by 8 m (heights in rollout_sites.csv).
- Aqaba: ASEZA permitting and customs regime; sectors facing Eilat/Taba carry $+2^\circ$ tilt and coordination limits.

RO-04 – Work Package

On-air window: Jan 2028 – Mar 2028 (M13–M15) · 719 sites · 2124 sectors · 16 hubs · 749 km access/aggregation fiber · 337 km backbone · 14 indoor venues

After this rollout: 2,107 sites on air (34 % of programme), $\approx$ 39.3 % of population covered, 376 km² of 1 Gbps footprint. Site list: filter `rollout = RO-04` in `rollout_sites.csv` / workbook sheet “Sites”; on the map choose “Rollout phase” and move the slider to 4.

1. Scope

Class	Sites	Site type	Sites
Dense urban	0	Rooftop	385
Urban	625	Monopole / greenfield	301
Suburban	61	Lattice tower	33
Rural	0		
Highway	33	Total	719

Areas: Amman-Zarqa Metro (466), Irbid (168), Aqaba (52), Desert Highway (R15) Amman-Aqaba (33).

Governorates: Amman 240, Zarqa 203, Irbid 168, Aqaba 60, Balqa 27, Ma'an 11, Karak 7, Tafilah 3.

2. Work packages (build units)

Unit	Type	Cluster / route	Sites
HUB-AMM-40	Hub cluster	Amman-Zarqa Metro	54
HUB-AMM-03	Hub cluster	Amman-Zarqa Metro	53
HUB-AQB-03	Hub cluster	Aqaba	52
HUB-AMM-07	Hub cluster	Amman-Zarqa Metro	48
HUB-IRB-11	Hub cluster	Irbid	44
HUB-IRB-17	Hub cluster	Irbid	42
HUB-AMM-01	Hub cluster	Amman-Zarqa Metro	42
HUB-AMM-29	Hub cluster	Amman-Zarqa Metro	42
HUB-AMM-24	Hub cluster	Amman-Zarqa Metro	42
HUB-IRB-08	Hub cluster	Irbid	41
HUB-IRB-07	Hub cluster	Irbid	41
HUB-AMM-39	Hub cluster	Amman-Zarqa Metro	41
HUB-AMM-15	Hub cluster	Amman-Zarqa Metro	41
HUB-AMM-54	Hub cluster	Amman-Zarqa Metro	38
HUB-AMM-51	Hub cluster	Amman-Zarqa Metro	36
HW	Desert Highway (R15) Amman-Aqaba	Highway corridor	Desert Highway (R15) Amman-Aqaba
HUB-AMM-36	Hub cluster	Amman-Zarqa Metro	29

3. Bill of quantities

Item	Qty	Item	Qty
n78 64T64R AAU	2058	Hubs / DU hotels	16

Item	Qty	Item	Qty
n78 8T8R (rural)	0	Hub aggregation routers	32
Passive 2L4H antenna + RET	2124	Cell-site routers 25GE	33
n28 RRU	2124	Access-ring fiber km	611
n1/n3 RRU	2124	Aggregation fiber km	108
Pooled DU (C-RAN, 0.75 per site)	515	Backbone add/drop spur km	30
Site DU (D-RAN)	33	Backbone / metro core km lit	337
Fronthaul gateways (C-RAN)	686		
n258 street cells	0	Microwave hops	0
Indoor pRRU / n258 heads	1545 / 47	Power systems (rectifier + battery)	719

4. Prerequisites (entry criteria)

- RO-03 gate G2 passed (teams and process release)
- Backbone lit in this rollout before G2: Dead Sea - Wadi Araba (R65) Amman-Aqaba (337 km)
- Backbone already in service from earlier rollouts: Amman metro core ring; Amman-Zarqa-Mafraq-Jaber; Desert Highway; Irbid-Jerash-Amman; Mafraq-Ramtha-Irbid
- Cross-border coordination filed for 1 flagged sectors in this rollout

5. Schedule

Activity	Start	End
Nominal release, TSSR surveys, site acquisition & permits	M7 · Jul 2027	M10 · Oct 2027
Detailed RF / transport design freeze, material call-off	M8 · Aug 2027	M10 · Oct 2027
Civil works, power, fiber build to hubs and rings	M9 · Sep 2027	M12 · Dec 2027
Hub / DU-hotel and router commissioning	M11 · Nov 2027	M13 · Jan 2028
Radio install, commissioning, integration to 5GC	M12 · Dec 2027	M15 · Mar 2028
Single-site verification (SSV)	M13 · Jan 2028	M15 · Mar 2028
Cluster optimisation & acceptance (1 Gbps drive test)	M14 · Feb 2028	M17 · May 2028
Commercial launch of rollout area / hand-over to operations	M16 · Apr 2028	M17 · May 2028

Required run-rate: $\approx$ 240 sites on air per month; $\approx$ 250 km of access/aggregation fiber per month.

6. Resources

20 acquisition agents · 181 civil crews · 63 urban fiber crews · 4 backbone crews · 35 radio install crews · 4 SSV teams · 5 optimisation teams.

7. Indoor venues in this rollout

Venue	Type	pRRU	n258
City Mall	mall	175	7
Mecca Mall	mall	208	8
King Hussein Business Park	campus	130	2
University of Jordan	campus	270	2
Jordan University Hospital	hospital	103	2

Venue	Type	pRRU	n258
King Hussein Cancer Center	hospital	119	2
Ayla Oasis / Hyatt Regency	resort	112	2
Al Manara - Saraya Aqaba	hotel	69	2
Kempinski Aqaba	hotel	64	2
InterContinental Aqaba	hotel	81	2
Movenpick Aqaba	hotel	90	2
Aqaba Gateway / City Center	mall	44	4
King Hussein Intl Airport (Aqaba)	airport	25	6
Arabella Mall Irbid	mall	55	4

8. Exit criteria

- G3: $\geq 95\%$ of sites on air, all access rings closed, no open critical alarms.
- G4: every cluster passes the acceptance KPIs of 03_Site_Implementation_Standard.md §6 ($\geq 95\%$ of bins ≥ 1 Gbps in DU/U, $\geq 93\%$ SU; rural/highway route KPIs).
- G5: as-built pack, asset register, spares and O&M hand-over signed; lessons-learned fed to RO-05.

9. Rollout-specific notes

- More than half the sites are rooftops – prioritise existing shared rooftops; antenna must clear local roofs by 8 m (heights in rollout_sites.csv).
- Aqaba: ASEZA permitting and customs regime; sectors facing Eilat/Taba carry $+2^\circ$ tilt and coordination limits.

RO-05 – Work Package

On-air window: Apr 2028 – Jun 2028 (M16–M18) · 769 sites · 2275 sectors · 19 hubs · 836 km access/aggregation fiber · 278 km backbone · 1 indoor venues

After this rollout: 2,876 sites on air (46 % of programme), ≈ 50.1 % of population covered, 550 km² of 1 Gbps footprint. Site list: filter `rollout = RO-05` in `rollout_sites.csv` / workbook sheet “Sites”; on the map choose “Rollout phase” and move the slider to 5.

1. Scope

Class	Sites	Site type	Sites
Dense urban	0	Rooftop	359
Urban	561	Monopole / greenfield	378
Suburban	176	Lattice tower	32
Rural	0		
Highway	32	Total	769

Areas: Amman-Zarqa Metro (618), Wadi Musa / Petra (65), Irbid (35), Dead Sea - Wadi Araba (R65) Amman-Aqaba (32), Dead Sea (19). **Governorates:** Amman 288, Zarqa 271, Balqa 71, Ma'an 65, Irbid 35, Aqaba 17, Madaba 11, Karak 8, Tafilah 3.

2. Work packages (build units)

Unit	Type	Cluster / route	Sites
HUB-AMM-33	Hub cluster	Amman-Zarqa Metro	52
HUB-AMM-70	Hub cluster	Amman-Zarqa Metro	48
HUB-AMM-46	Hub cluster	Amman-Zarqa Metro	48
HUB-AMM-86	Hub cluster	Amman-Zarqa Metro	47
HUB-AMM-62	Hub cluster	Amman-Zarqa Metro	46
HUB-AMM-66	Hub cluster	Amman-Zarqa Metro	42
HUB-AMM-77	Hub cluster	Amman-Zarqa Metro	41
HUB-AMM-19	Hub cluster	Amman-Zarqa Metro	41
HUB-AMM-61	Hub cluster	Amman-Zarqa Metro	39
HUB-AMM-72	Hub cluster	Amman-Zarqa Metro	39
HUB-AMM-65	Hub cluster	Amman-Zarqa Metro	37
HUB-AMM-22	Hub cluster	Amman-Zarqa Metro	37
HUB-AMM-85	Hub cluster	Amman-Zarqa Metro	35
HUB-IRB-03	Hub cluster	Irbid	35
HUB-WAD-01	Hub cluster	Wadi Musa / Petra	34
HUB-AMM-63	Hub cluster	Amman-Zarqa Metro	33
HUB-AMM-42	Hub cluster	Amman-Zarqa Metro	33
HW	Dead Sea - Wadi Araba (R65) Amman-Aqaba	Highway corridor	Dead Sea - Wadi Araba (R65) Amman-Aqaba
HUB-WAD-02	Hub cluster	Wadi Musa / Petra	31
HUB-DEA-01	Hub cluster	Dead Sea	19

3. Bill of quantities

Item	Qty	Item	Qty
n78 64T64R AAU	2211	Hubs / DU hotels	19
n78 8T8R (rural)	0	Hub aggregation routers	38
Passive 2L4H antenna + RET	2275	Cell-site routers 25GE	85
n28 RRU	2275	Access-ring fiber km	697
n1/n3 RRU	2275	Aggregation fiber km	109
Pooled DU (C-RAN, 0.75 per site)	513	Backbone add/drop spur km	30
Site DU (D-RAN)	85	Backbone / metro core km lit	278
Fronthaul gateways (C-RAN)	684		
n258 street cells	0	Microwave hops	0
Indoor pRRU / n258 heads	88 / 4	Power systems (rectifier + battery)	769

4. Prerequisites (entry criteria)

- RO-04 gate G2 passed (teams and process release)
- Backbone lit in this rollout before G2: King's Highway Madaba-Karak-Tafilah-Petra-Ma'an (241 km); Amman-Salt-Deir Alla (37 km)
- Backbone already in service from earlier rollouts: Amman metro core ring; Amman-Zarqa-Mafraq-Jaber; Dead Sea - Wadi Araba; Desert Highway; Irbid-Jerash-Amman; Mafraq-Ramtha-Irbid

5. Schedule

Activity	Start	End
Nominal release, TSSR surveys, site acquisition & permits	M10 · Oct 2027	M13 · Jan 2028
Detailed RF / transport design freeze, material call-off	M11 · Nov 2027	M13 · Jan 2028
Civil works, power, fiber build to hubs and rings	M12 · Dec 2027	M15 · Mar 2028
Hub / DU-hotel and router commissioning	M14 · Feb 2028	M16 · Apr 2028
Radio install, commissioning, integration to 5GC	M15 · Mar 2028	M18 · Jun 2028
Single-site verification (SSV)	M16 · Apr 2028	M18 · Jun 2028
Cluster optimisation & acceptance (1 Gbps drive test)	M17 · May 2028	M20 · Aug 2028
Commercial launch of rollout area / hand-over to operations	M19 · Jul 2028	M20 · Aug 2028

Required run-rate: $\approx$ 256 sites on air per month; $\approx$ 279 km of access/aggregation fiber per month.

6. Resources

22 acquisition agents · 202 civil crews · 70 urban fiber crews · 3 backbone crews · 37 radio install crews · 5 SSV teams · 6 optimisation teams.

7. Indoor venues in this rollout

Venue	Type	pRRU	n258
Irbid City Centre	mall	88	4

8. Exit criteria

- G3: $\geq 95\%$ of sites on air, all access rings closed, no open critical alarms.
- G4: every cluster passes the acceptance KPIs of 03_Site_Implementation_Standard.md §6 ($\geq 95\%$ of bins ≥ 1 Gbps in DU/U, $\geq 93\%$ SU; rural/highway route KPIs).
- G5: as-built pack, asset register, spares and O&M hand-over signed; lessons-learned fed to RO-06.

9. Rollout-specific notes

- Aqaba: ASEZA permitting and customs regime; sectors facing Eilat/Taba carry $+2^\circ$ tilt and coordination limits.
- Tourism zones (Dead Sea / Petra): camouflage structures, heritage-authority approval (PDTRA for Petra), work outside peak season where possible.

RO-06 – Work Package

On-air window: Jul 2028 – Sep 2028 (M19–M21) · 770 sites · 2303 sectors · 19 hubs · 922 km access/aggregation fiber · 174 km backbone · 9 indoor venues

After this rollout: 3,646 sites on air (58 % of programme), $\approx$ 60.3 % of population covered, 755 km² of 1 Gbps footprint. Site list: filter `rollout = RO-06` in `rollout_sites.csv` / workbook sheet “Sites”; on the map choose “Rollout phase” and move the slider to 6.

1. Scope

Class	Sites	Site type	Sites
Dense urban	0	Rooftop	225
Urban	455	Monopole / greenfield	538
Suburban	308	Lattice tower	7
Rural	0		
Highway	7	Total	770

Areas: Irbid (283), Amman-Zarqa Metro (144), Madaba (122), Mafraq (75), Tafilah (42), Karak (38), Ma'an (35), Aqaba (24), King's Highway Madaba-Karak-Tafilah-Petra-Ma'an (7). **Governorates:** Irbid 283, Amman 131, Madaba 87, Mafraq 75, Zarqa 48, Tafilah 43, Ma'an 41, Karak 38, Aqaba 24.

2. Work packages (build units)

Unit	Type	Cluster / route	Sites
HUB-IRB-13	Hub cluster	Irbid	71
HUB-IRB-06	Hub cluster	Irbid	60
HUB-IRB-18	Hub cluster	Irbid	42
HUB-TAF-02	Hub cluster	Tafilah	42
HUB-MAD-04	Hub cluster	Madaba	42
HUB-AMM-81	Hub cluster	Amman-Zarqa Metro	41
HUB-MAD-02	Hub cluster	Madaba	41
HUB-MAF-05	Hub cluster	Mafraq	39
HUB-MAD-03	Hub cluster	Madaba	39
HUB-IRB-04	Hub cluster	Irbid	38
HUB-KAR-03	Hub cluster	Karak	38
HUB-IRB-09	Hub cluster	Irbid	37
HUB-AMM-60	Hub cluster	Amman-Zarqa Metro	37
HUB-MAF-02	Hub cluster	Mafraq	36
HUB-MA-02	Hub cluster	Ma'an	35
HUB-IRB-19	Hub cluster	Irbid	35

Unit	Type	Cluster / route	Sites
HUB-AMM-76	Hub cluster	Amman-Zarqa Metro	33
HUB-AMM-06	Hub cluster	Amman-Zarqa Metro	33
HUB-AQB-02	Hub cluster	Aqaba	24
HW	King's Highway Madaba-Karak-Tafilah-Petra-Ma'an	Highway corridor	King's Highway Madaba-Karak-Tafilah-Petra-Ma'an

3. Bill of quantities

Item	Qty	Item	Qty
n78 64T64R AAU	2289	Hubs / DU hotels	19
n78 8T8R (rural)	0	Hub aggregation routers	38
Passive 2L4H antenna + RET	2303	Cell-site routers 25GE	173
n28 RRU	2303	Access-ring fiber km	810
n1/n3 RRU	2303	Aggregation fiber km	106
Pooled DU (C-RAN, 0.75 per site)	448	Backbone add/drop spur km	6
Site DU (D-RAN)	173	Backbone / metro core km lit	174
Fronthaul gateways (C-RAN)	597		
n258 street cells	0	Microwave hops	0
Indoor pRRU / n258 heads	650 / 28	Power systems (rectifier + battery)	770

4. Prerequisites (entry criteria)

- RO-05 gate G2 passed (teams and process release)
- Backbone lit in this rollout before G2: Jordan Valley (R65 north) (127 km); Irbid-Ajloun-Jerash (47 km)
- Backbone already in service from earlier rollouts: Amman metro core ring; Amman-Salt-Deir Alla; Amman-Zarqa-Mafraq-Jaber; Dead Sea - Wadi Araba; Desert Highway; Irbid-Jerash-Amman; King's Highway Madaba-Karak-Tafilah-Petra-Ma'an; Mafraq-Ramtha-Irbid

5. Schedule

Activity	Start	End
Nominal release, TSSR surveys, site acquisition & permits	M13 · Jan 2028	M16 · Apr 2028
Detailed RF / transport design freeze, material call-off	M14 · Feb 2028	M16 · Apr 2028
Civil works, power, fiber build to hubs and rings	M15 · Mar 2028	M18 · Jun 2028
Hub / DU-hotel and router commissioning	M17 · May 2028	M19 · Jul 2028
Radio install, commissioning, integration to 5GC	M18 · Jun 2028	M21 · Sep 2028
Single-site verification (SSV)	M19 · Jul 2028	M21 · Sep 2028
Cluster optimisation & acceptance (1 Gbps drive test)	M20 · Aug 2028	M23 · Nov 2028
Commercial launch of rollout area / hand-over to operations	M22 · Oct 2028	M23 · Nov 2028

Required run-rate: ≈ 257 sites on air per month; ≈ 308 km of access/aggregation fiber per month.

6. Resources

22 acquisition agents · 221 civil crews · 77 urban fiber crews · 2 backbone crews · 37 radio install crews · 5 SSV teams · 6 optimisation teams.

7. Indoor venues in this rollout

Venue	Type	pRRU	n258
Kempinski Ishtar Dead Sea	hotel	111	4
Movenpick Dead Sea	hotel	109	4
Dead Sea Marriott	hotel	80	2
Hilton Dead Sea	hotel	90	2
Crowne Plaza Dead Sea	hotel	126	4
King Hussein Bin Talal Convention Centre	convention	42	6
Movenpick Petra	hotel	52	2
Petra Marriott	hotel	31	2
Petra Visitor Centre	venue	9	2

8. Exit criteria

- G3: $\geq 95\%$ of sites on air, all access rings closed, no open critical alarms.
- G4: every cluster passes the acceptance KPIs of 03_Site_Implementation_Standard.md §6 ($\geq 95\%$ of bins ≥ 1 Gbps in DU/U, $\geq 93\%$ SU; rural/highway route KPIs).
- G5: as-built pack, asset register, spares and O&M hand-over signed; lessons-learned fed to RO-07.

9. Rollout-specific notes

- Standard process applies.

RO-07 – Work Package

On-air window: Oct 2028 – Dec 2028 (M22–M24) · 766 sites · 2298 sectors · 13 hubs · 576 km access/aggregation fiber · 185 km backbone · 0 indoor venues

After this rollout: 4,412 sites on air (70 % of programme), ≈ 73.9 % of population covered, 878 km² of 1 Gbps footprint. Site list: filter `rollout = RO-07` in `rollout_sites.csv` / workbook sheet “Sites”; on the map choose “Rollout phase” and move the slider to 7.

1. Scope

Class	Sites	Site type	Sites
Dense urban	0	Rooftop	132
Urban	264	Monopole / greenfield	323
Suburban	191	Lattice tower	311
Rural	311		
Highway	0	Total	766

Areas: Jerash (117), Ma'an (102), Salt (101), Mafraq (71), Karak governorate (rural) (40), Amman governorate (rural) (40), Madaba governorate (rural) (40), Mafraq governorate (rural) (40), Irbid governorate (rural) (40), Tafilah governorate (rural) (35), Karak (35), Amman-Zarqa Metro (29), Balqa governorate (rural) (28), Ajloun governorate (rural) (19), Jerash governorate (rural) (16), Zarqa governorate (rural) (13). **Governorates:** Balqa 158, Jerash 133, Mafraq 111, Ma'an 102, Karak 75, Madaba 40, Amman 40, Irbid 40, Tafilah 35, Ajloun 19, Zarqa 13.

2. Work packages (build units)

Unit	Type	Cluster / route	Sites
RUR	Karak	1	Rural package
RUR	Amman	1	Rural package
RUR	Madaba	1	Rural package
RUR	Mafraq	1	Rural package
RUR	Irbid	1	Rural package
HUB-JER-02	Hub cluster	Jerash	40
HUB-JER-03	Hub cluster	Jerash	39
HUB-JER-01	Hub cluster	Jerash	38
HUB-MAF-03	Hub cluster	Mafraq	36
RUR	Tafilah	1	Rural package
HUB-MA-03	Hub cluster	Ma'an	35
HUB-SAL-01	Hub cluster	Salt	35
HUB-SAL-02	Hub cluster	Salt	35
HUB-MA-01	Hub cluster	Ma'an	35
HUB-MAF-04	Hub cluster	Mafraq	35
HUB-KAR-02	Hub cluster	Karak	35
HUB-MA-04	Hub cluster	Ma'an	32
HUB-SAL-03	Hub cluster	Salt	31
HUB-AMM-71	Hub cluster	Amman-Zarqa Metro	29

Unit	Type	Cluster / route	Sites
RUR	Balqa	1	Rural package
RUR	Ajloun	1	Rural package
RUR	Jerash	1	Rural package
RUR	Zarqa	1	Rural package

3. Bill of quantities

Item	Qty	Item	Qty
n78 64T64R AAU	1365	Hubs / DU hotels	13
n78 8T8R (rural)	933	Hub aggregation routers	26
Passive 2L4H antenna + RET	2298	Cell-site routers 25GE	340
n28 RRU	2298	Access-ring fiber km	463
n1/n3 RRU	2298	Aggregation fiber km	53
Pooled DU (C-RAN, 0.75 per site)	320	Backbone add/drop spur km	60
Site DU (D-RAN)	340	Backbone / metro core km lit	185
Fronthaul gateways (C-RAN)	426		
n258 street cells	0	Microwave hops	246
Indoor pRRU / n258 heads	0 / 0	Power systems (rectifier + battery)	766

4. Prerequisites (entry criteria)

- RO-06 gate G2 passed (teams and process release)
- Backbone lit in this rollout before G2: Karak-Qatranah (39 km); Tafilah-Jurf (34 km); Amman-Azraq (R40) (112 km)
- Backbone already in service from earlier rollouts: Amman metro core ring; Amman-Salt-Deir Alla; Amman-Zarqa-Mafraq-Jaber; Dead Sea - Wadi Araba; Desert Highway; Irbid-Ajloun-Jerash; Irbid-Jerash-Amman; Jordan Valley ...

5. Schedule

Activity	Start	End
Nominal release, TSSR surveys, site acquisition & permits	M16 · Apr 2028	M19 · Jul 2028
Detailed RF / transport design freeze, material call-off	M17 · May 2028	M19 · Jul 2028
Civil works, power, fiber build to hubs and rings	M18 · Jun 2028	M21 · Sep 2028
Hub / DU-hotel and router commissioning	M20 · Aug 2028	M22 · Oct 2028
Radio install, commissioning, integration to 5GC	M21 · Sep 2028	M24 · Dec 2028
Single-site verification (SSV)	M22 · Oct 2028	M24 · Dec 2028
Cluster optimisation & acceptance (1 Gbps drive test)	M23 · Nov 2028	M26 · Feb 2029
Commercial launch of rollout area / hand-over to operations	M25 · Jan 2029	M26 · Feb 2029

Required run-rate: ≈ 255 sites on air per month; ≈ 192 km of access/aggregation fiber per month.

6. Resources

22 acquisition agents · 278 civil crews · 48 urban fiber crews · 2 backbone crews · 37 radio install crews · 5 SSV teams · 6 optimisation teams.

7. Indoor venues in this rollout

None.

8. Exit criteria

- G3: ≥ 95 % of sites on air, all access rings closed, no open critical alarms.
- G4: every cluster passes the acceptance KPIs of 03_Site_Implementation_Standard.md §6 (≥ 95 % of bins ≥ 1 Gbps in DU/U, ≥ 93 % SU; rural/highway route KPIs).
- G5: as-built pack, asset register, spares and O&M hand-over signed; lessons-learned fed to RO-08.

9. Rollout-specific notes

- Rural/highway towers: 45 m lattice, solar-hybrid power where off-grid, backbone add/drop or microwave per backhaul column; security fencing standard.

RO-08 – Work Package

On-air window: Jan 2029 – Mar 2029 (M25–M27) · 718 sites · 2142 sectors · 18 hubs · 802 km access/aggregation fiber · 471 km backbone · 2 indoor venues

After this rollout: 5,130 sites on air (82 % of programme), ≈ 83.4 % of population covered, 1,071 km² of 1 Gbps footprint. Site list: filter `rollout = RO-08` in `rollout_sites.csv` / workbook sheet “Sites”; on the map choose “Rollout phase” and move the slider to 8.

1. Scope

Class	Sites	Site type	Sites
Dense urban	0	Rooftop	181
Urban	166	Monopole / greenfield	400
Suburban	415	Lattice tower	137
Rural	126		
Highway	11	Total	718

Areas: Amman-Zarqa Metro (288), Ajloun (81), Amman governorate (rural) (40), Salt (36), Madaba (36), Karak governorate (rural) (35), Karak (34), Mafrq (31), Tafilah (30), Mafrq governorate (rural) (29), Irbid (27), Irbid governorate (rural) (21), Aqaba (18), Amman-Azraq (R40) (9), Karak-Qatraneh (1), Tafilah-Jurf (1), Madaba governorate (rural) (1). **Governorates:** Amman 212, Zarqa 105, Ajloun 81, Karak 70, Mafrq 60, Balqa 58, Irbid 48, Madaba 35, Tafilah 31, Aqaba 18.

2. Work packages (build units)

Unit	Type	Cluster / route	Sites
HUB-AMM-37	Hub cluster	Amman-Zarqa Metro	48
HUB-AMM-73	Hub cluster	Amman-Zarqa Metro	46
RUR	Amman	2	Rural package
HUB-SAL-04	Hub cluster	Salt	36
HUB-MAD-01	Hub cluster	Madaba	36
HUB-AMM-58	Hub cluster	Amman-Zarqa Metro	36
RUR	Karak	2	Rural package
HUB-AMM-21	Hub cluster	Amman-Zarqa Metro	35
HUB-KAR-01	Hub cluster	Karak	34
HUB-AMM-50	Hub cluster	Amman-Zarqa Metro	34
HUB-AMM-20	Hub cluster	Amman-Zarqa Metro	33
HUB-MAF-01	Hub cluster	Mafrq	31
HUB-TAF-01	Hub cluster	Tafilah	30
HUB-AMM-84	Hub cluster	Amman-Zarqa Metro	30
RUR	Mafrq	2	Rural package
HUB-AJL-02	Hub cluster	Ajloun	29
HUB-AJL-03	Hub cluster	Ajloun	27
HUB-IRB-12	Hub cluster	Irbid	27
HUB-AMM-74	Hub cluster	Amman-Zarqa Metro	26

Unit	Type	Cluster / route	Sites
HUB-AJL-01	Hub cluster	Ajloun	25
RUR	Irbid	2	Rural package
HUB-AQB-04	Hub cluster	Aqaba	18
HW	Amman-Azraq (R40)	Highway corridor	Amman-Azraq (R40)
HW	Karak-Qatraneh	Highway corridor	Karak-Qatraneh
HW	Tafilah-Jurf	Highway corridor	Tafilah-Jurf
RUR	Madaba	2	Rural package

3. Bill of quantities

Item	Qty	Item	Qty
n78 64T64R AAU	1743	Hubs / DU hotels	18
n78 8T8R (rural)	377	Hub aggregation routers	36
Passive 2L4H antenna + RET	2142	Cell-site routers 25GE	586
n28 RRU	2142	Access-ring fiber km	667
n1/n3 RRU	2142	Aggregation fiber km	107
Pooled DU (C-RAN, 0.75 per site)	99	Backbone add/drop spur km	28
Site DU (D-RAN)	586	Backbone / metro core km lit	471
Fronthaul gateways (C-RAN)	132		
n258 street cells	0	Microwave hops	107
Indoor pRRU / n258 heads	319 / 4	Power systems (rectifier + battery)	718

4. Prerequisites (entry criteria)

- RO-07 gate G2 passed (teams and process release)
- Backbone lit in this rollout before G2: Zarqa-Azraq-Safawi (137 km); Mafraq-Safawi-Ruwaished (R10) (302 km); Quweira-Wadi Rum (32 km)
- Backbone already in service from earlier rollouts: Amman metro core ring; Amman-Azraq; Amman-Salt-Deir Alla; Amman-Zarqa-Mafraq-Jaber; Dead Sea - Wadi Araba; Desert Highway; Irbid-Ajloun-Jerash; Irbid-Jerash-Amman ...
- Cross-border coordination filed for 17 flagged sectors in this rollout

5. Schedule

Activity	Start	End
Nominal release, TSSR surveys, site acquisition & permits	M19 · Jul 2028	M22 · Oct 2028
Detailed RF / transport design freeze, material call-off	M20 · Aug 2028	M22 · Oct 2028
Civil works, power, fiber build to hubs and rings	M21 · Sep 2028	M24 · Dec 2028
Hub / DU-hotel and router commissioning	M23 · Nov 2028	M25 · Jan 2029
Radio install, commissioning, integration to 5GC	M24 · Dec 2028	M27 · Mar 2029
Single-site verification (SSV)	M25 · Jan 2029	M27 · Mar 2029
Cluster optimisation & acceptance (1 Gbps drive test)	M26 · Feb 2029	M29 · May 2029
Commercial launch of rollout area / hand-over to operations	M28 · Apr 2029	M29 · May 2029

Required run-rate: $\approx$ 239 sites on air per month; $\approx$ 268 km of access/aggregation fiber per month.

6. Resources

20 acquisition agents · 229 civil crews · 67 urban fiber crews · 5 backbone crews · 35 radio install crews · 4 SSV teams · 5 optimisation teams.

7. Indoor venues in this rollout

Venue	Type	pRRU	n258
JUST campus	campus	216	2
King Abdullah University Hospital	hospital	103	2

8. Exit criteria

- G3: $\geq 95\%$ of sites on air, all access rings closed, no open critical alarms.
- G4: every cluster passes the acceptance KPIs of `03_Site_Implementation_Standard.md` §6 ($\geq 95\%$ of bins ≥ 1 Gbps in DU/U, $\geq 93\%$ SU; rural/highway route KPIs).
- G5: as-built pack, asset register, spares and O&M hand-over signed; lessons-learned fed to RO-09.

9. Rollout-specific notes

- Rural/highway towers: 45 m lattice, solar-hybrid power where off-grid, backbone add/drop or microwave per `backhaul` column; security fencing standard.

RO-09 – Work Package

On-air window: Apr 2029 – Jun 2029 (M28–M30) · 669 sites · 1956 sectors · 17 hubs · 932 km access/aggregation fiber · 0 km backbone · 0 indoor venues

After this rollout: 5,799 sites on air (92 % of programme), ≈ 88.9 % of population covered, 1,309 km² of 1 Gbps footprint. Site list: filter `rollout = RO-09` in `rollout_sites.csv` / workbook sheet “Sites”; on the map choose “Rollout phase” and move the slider to 9.

1. Scope

Class	Sites	Site type	Sites
Dense urban	0	Rooftop	83
Urban	2	Monopole / greenfield	528
Suburban	609	Lattice tower	58
Rural	7		
Highway	51	Total	669

Areas: Amman-Zarqa Metro (550), Irbid (61), Mafrq-Safawi-Ruwaished (R10) (34), Zarqa-Azraq-Safawi (13), Amman governorate (rural) (7), Quweira-Wadi Rum (4). **Governorates:** Amman 361, Zarqa 133, Balqa 71, Irbid 61, Mafrq 39, Aqaba 4.

2. Work packages (build units)

Unit	Type	Cluster / route	Sites
HUB-AMM-78	Hub cluster	Amman-Zarqa Metro	55
HUB-AMM-16	Hub cluster	Amman-Zarqa Metro	49
HUB-AMM-30	Hub cluster	Amman-Zarqa Metro	47
HUB-AMM-32	Hub cluster	Amman-Zarqa Metro	42
HUB-AMM-55	Hub cluster	Amman-Zarqa Metro	40
HUB-AMM-25	Hub cluster	Amman-Zarqa Metro	39
HUB-AMM-53	Hub cluster	Amman-Zarqa Metro	37
HUB-AMM-34	Hub cluster	Amman-Zarqa Metro	36
HUB-AMM-43	Hub cluster	Amman-Zarqa Metro	36
HUB-AMM-31	Hub cluster	Amman-Zarqa Metro	35
HW	Mafrq-Safawi-Ruwaished (R10)	Highway corridor	Mafrq-Safawi-Ruwaished (R10)
HUB-IRB-16	Hub cluster	Irbid	33
HUB-AMM-68	Hub cluster	Amman-Zarqa Metro	33
HUB-AMM-67	Hub cluster	Amman-Zarqa Metro	32
HUB-AMM-10	Hub cluster	Amman-Zarqa Metro	28
HUB-IRB-02	Hub cluster	Irbid	28
HUB-AMM-26	Hub cluster	Amman-Zarqa Metro	22
HUB-AMM-18	Hub cluster	Amman-Zarqa Metro	19
HW	Zarqa-Azraq-Safawi	Highway corridor	Zarqa-Azraq-Safawi
RUR	Amman	3	Rural package

Unit	Type	Cluster / route	Sites
HW	Quweira-Wadi Rum	Highway corridor	Quweira-Wadi Rum

3. Bill of quantities

Item	Qty	Item	Qty
n78 64T64R AAU	1833	Hubs / DU hotels	17
n78 8T8R (rural)	21	Hub aggregation routers	34
Passive 2L4H antenna + RET	1956	Cell-site routers 25GE	669
n28 RRU	1956	Access-ring fiber km	764
n1/n3 RRU	1956	Aggregation fiber km	121
Pooled DU (C-RAN, 0.75 per site)	0	Backbone add/drop spur km	47
Site DU (D-RAN)	669	Backbone / metro core km lit	0
Fronthaul gateways (C-RAN)	0		
n258 street cells	480	Microwave hops	7
Indoor pRRU / n258 heads	0 / 0	Power systems (rectifier + battery)	669

4. Prerequisites (entry criteria)

- RO-08 gate G2 passed (teams and process release)
- Backbone already in service from earlier rollouts: Amman metro core ring; Amman-Azraq; Amman-Salt-Deir Alla; Amman-Zarqa-Mafraq-Jaber; Dead Sea - Wadi Araba; Desert Highway; Irbid-Ajloun-Jerash; Irbid-Jerash-Amman ...

5. Schedule

Activity	Start	End
Nominal release, TSSR surveys, site acquisition & permits	M22 · Oct 2028	M25 · Jan 2029
Detailed RF / transport design freeze, material call-off	M23 · Nov 2028	M25 · Jan 2029
Civil works, power, fiber build to hubs and rings	M24 · Dec 2028	M27 · Mar 2029
Hub / DU-hotel and router commissioning	M26 · Feb 2029	M28 · Apr 2029
Radio install, commissioning, integration to 5GC	M27 · Mar 2029	M30 · Jun 2029
Single-site verification (SSV)	M28 · Apr 2029	M30 · Jun 2029
Cluster optimisation & acceptance (1 Gbps drive test)	M29 · May 2029	M32 · Aug 2029
Commercial launch of rollout area / hand-over to operations	M31 · Jul 2029	M32 · Aug 2029

Required run-rate: ≈ 223 sites on air per month; ≈ 311 km of access/aggregation fiber per month.

6. Resources

19 acquisition agents · 218 civil crews · 78 urban fiber crews · 0 backbone crews · 32 radio install crews · 4 SSV teams · 5 optimisation teams.

7. Indoor venues in this rollout

None.

8. Exit criteria

- G3: ≥ 95 % of sites on air, all access rings closed, no open critical alarms.
- G4: every cluster passes the acceptance KPIs of 03_Site_Implementation_Standard.md §6 (≥ 95 % of bins ≥ 1 Gbps in DU/U, ≥ 93 % SU; rural/highway route KPIs).
- G5: as-built pack, asset register, spares and O&M hand-over signed; lessons-learned fed to RO-10.

9. Rollout-specific notes

- Rural/highway towers: 45 m lattice, solar-hybrid power where off-grid, backbone add/drop or microwave per backhaul column; security fencing standard.
- Includes 480 n258 street small cells for dense-urban hotspots (municipal street-furniture agreement required).

RO-10 – Work Package

On-air window: Jul 2029 – Sep 2029 (M31–M33) · 471 sites · 1413 sectors · 17 hubs · 659 km access/aggregation fiber · 0 km backbone · 2 indoor venues

After this rollout: 6,270 sites on air (100 % of programme), ≈ 93.0 % of population covered, 1,491 km² of 1 Gbps footprint. Site list: filter `rollout = RO-10` in `rollout_sites.csv` / workbook sheet “Sites”; on the map choose “Rollout phase” and move the slider to 10.

1. Scope

Class	Sites	Site type	Sites
Dense urban	0	Rooftop	42
Urban	6	Monopole / greenfield	429
Suburban	465	Lattice tower	0
Rural	0		
Highway	0	Total	471

Areas: Amman-Zarqa Metro (208), Irbid (71), Mutah-Mazar (36), Aqaba (34), Azraq (32), Deir Alla (24), Ghor Safi (23), South Shuna (23), North Shuna (20). **Governorates:** Balqa 113, Zarqa 110, Irbid 94, Karak 59, Amman 58, Aqaba 34, Ajloun 3.

2. Work packages (build units)

Unit	Type	Cluster / route	Sites
HUB-AMM-80	Hub cluster	Amman-Zarqa Metro	50
HUB-AMM-75	Hub cluster	Amman-Zarqa Metro	37
HUB-MUT-01	Hub cluster	Mutah-Mazar	36
HUB-AMM-48	Hub cluster	Amman-Zarqa Metro	35
HUB-AQB-01	Hub cluster	Aqaba	34
HUB-AZR-01	Hub cluster	Azraq	32
HUB-IRB-20	Hub cluster	Irbid	26
HUB-IRB-14	Hub cluster	Irbid	24
HUB-AMM-08	Hub cluster	Amman-Zarqa Metro	24
HUB-DEI-01	Hub cluster	Deir Alla	24
HUB-GHO-01	Hub cluster	Ghor Safi	23
HUB-SOU-01	Hub cluster	South Shuna	23
HUB-AMM-17	Hub cluster	Amman-Zarqa Metro	22
HUB-AMM-12	Hub cluster	Amman-Zarqa Metro	21
HUB-IRB-10	Hub cluster	Irbid	21
HUB-NOR-01	Hub cluster	North Shuna	20
HUB-AMM-09	Hub cluster	Amman-Zarqa Metro	19

3. Bill of quantities

Item	Qty	Item	Qty
n78 64T64R AAU	1413	Hubs / DU hotels	17
n78 8T8R (rural)	0	Hub aggregation routers	34
Passive 2L4H antenna + RET	1413	Cell-site routers 25GE	471
n28 RRU	1413	Access-ring fiber km	579
n1/n3 RRU	1413	Aggregation fiber km	80
Pooled DU (C-RAN, 0.75 per site)	0	Backbone add/drop spur km	0
Site DU (D-RAN)	471	Backbone / metro core km lit	0
Fronthaul gateways (C-RAN)	0		
n258 street cells	480	Microwave hops	0
Indoor pRRU / n258 heads	424 / 17	Power systems (rectifier + battery)	471

4. Prerequisites (entry criteria)

- RO-09 gate G2 passed (teams and process release)
- Backbone already in service from earlier rollouts: Amman metro core ring; Amman-Azraq; Amman-Salt-Deir Alla; Amman-Zarqa-Mafraq-Jaber; Dead Sea - Wadi Araba; Desert Highway; Irbid-Ajloun-Jerash; Irbid-Jerash-Amman ...
- Cross-border coordination filed for 27 flagged sectors in this rollout

5. Schedule

Activity	Start	End
Nominal release, TSSR surveys, site acquisition & permits	M25 · Jan 2029	M28 · Apr 2029
Detailed RF / transport design freeze, material call-off	M26 · Feb 2029	M28 · Apr 2029
Civil works, power, fiber build to hubs and rings	M27 · Mar 2029	M30 · Jun 2029
Hub / DU-hotel and router commissioning	M29 · May 2029	M31 · Jul 2029
Radio install, commissioning, integration to 5GC	M30 · Jun 2029	M33 · Sep 2029
Single-site verification (SSV)	M31 · Jul 2029	M33 · Sep 2029
Cluster optimisation & acceptance (1 Gbps drive test)	M32 · Aug 2029	M35 · Nov 2029
Commercial launch of rollout area / hand-over to operations	M34 · Oct 2029	M35 · Nov 2029

Required run-rate: $\approx$ 157 sites on air per month; $\approx$ 220 km of access/aggregation fiber per month.

6. Resources

14 acquisition agents · 150 civil crews · 55 urban fiber crews · 0 backbone crews · 23 radio install crews · 3 SSV teams · 4 optimisation teams.

7. Indoor venues in this rollout

Venue	Type	pRRU	n258
Queen Alia Intl Airport terminal	airport	172	13
Tala Bay resort cluster	resort	252	4

8. Exit criteria

- G3: $\geq 95\%$ of sites on air, all access rings closed, no open critical alarms.
- G4: every cluster passes the acceptance KPIs of 03_Site_Implementation_Standard.md §6 ($\geq 95\%$ of bins ≥ 1 Gbps in DU/U, $\geq 93\%$ SU; rural/highway route KPIs).
- G5: as-built pack, asset register, spares and O&M hand-over signed; lessons-learned fed to operations.

9. Rollout-specific notes

- Aqaba: ASEZA permitting and customs regime; sectors facing Eilat/Taba carry $+2^\circ$ tilt and coordination limits.
- Includes 480 n258 street small cells for dense-urban hotspots (municipal street-furniture agreement required).
- Final rollout: network-wide parameter audit, neighbour clean-up, as-built documentation and O&M hand-over run in parallel (M34–M36).